Journal of
Theoretical and Applied Electronic Commerce Research

MDPI

Article

From Connectivity to Commerce: A Multi-Technique Investigation of E-Commerce Drivers in Italy's Regional Landscape

Angelo Leogrande [1,*], Carlo Drago [2], Alberto Costantiello [1] and Massimo Arnone [3]

[1] Department of Management, Finance and Technology, LUM University Giuseppe Degennaro, 70100 Casamassima, Italy; costantiello@lum.it
[2] Department of Economics, Psychology, Communication, Education, and Motor Sciences, Niccolò Cusano University, 00166 Rome, Italy; carlo.drago@unicusano.it
[3] Department of Political and Social Sciences, University of Catania, 95131 Catania, Italy; massimo.arnone@unict.it
* Correspondence: leogrande.cultore@lum.it

Abstract

The research examines regional disparities in the diffusion of e-commerce among enterprises employing at least 10 people in Italy, using an integrated analytical framework that blends econometric modeling, machine learning, and network analysis. Instrumental Variable (IV) panel models overcome endogeneity arising from digital infrastructure, socioeconomic factors, and online business activity, with geographic slope as a suitable instrument for broadband penetration. Machine learning models—regularized regressions, random forests, and boosting—augment causal inference by registering nonlinear effects and sorting variable salience. The results, in all cases, emphasize internet use, household digital connectivity, and the prevalence of remote work as the most important predictors of the diffusion of e-commerce. Cluster analysis identifies regional digital profiles that distinguish northern-central regions from southern-insular regions, characterizing persistently distinct digital divides. The network analysis, in turn, identifies digital inclusion variables—such as internet penetration and ICT infrastructure—that occupy central positions within the entirety of the economic and technological interdependencies' regime. Innovation and income levels, while practiced, hold peripheral positions, indicating that digital capacity, rather than economic affluence in the singular, drives online business participation. Italy's case can particularly illustrate this beyond its national borders. Being a high-income economy with significant regional disparities, it reproduces challenges common elsewhere in the world, among which the cases of Spain, Germany, the USA, the Republic of Korea, and Japan come to mind, where regional disparities inhibit inclusive digital development. The Italian case presents, then, a transferable model for the diffusion of digital tools, the reduction in regional disparities, and the encouragement of economic integration. By synthesizing the causal, predictive, and systemic methodologies, the study offers a theoretical and practical response to digital transformation across diverse terrains.

Keywords: e-commerce; regional disparities; digital transformation; machine learning; network analysis

JEL Classification: O33; R11; L81; C23; D83

Check for updates

Academic Editors: Ruiliang Yan and Zhi Pei

Received: 16 March 2026
Revised: 21 April 2026
Accepted: 22 April 2026
Published: 28 April 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

E-commerce diffusion represents an important aspect of digital transformation processes that redefine the way firms interact with their customers, reshape logistics networks, and modify the nature of competition in digitally driven economies. While major multinational companies have been the first to adopt innovations, SMEs are now playing an ever-increasing role in this regard. Nevertheless, the diffusion of e-commerce varies significantly from one region to another due to differences in infrastructure, employment opportunities, and institutional development. Thus, there is an urgent need to uncover the main factors influencing digital transformation and the reasons behind regional divergence. As demonstrated by numerous scholarly works, the issue of technological readiness and skills can explain why some regions and businesses adopt digital technologies and others reject them. However, such research typically focuses on static or cross-sectional patterns without accounting for spatial heterogeneity and underlying mechanisms [1–4]. Italy-focused studies also appear to be descriptive and lacking the insights into how different technological, socio-economic, and institutional factors interact [5,6]. Recently, however, more scholars pay attention to the increasing role of sophisticated digital solutions in the context of the e-commerce sector [7,8]. Some other works demonstrate that the peculiarities of digital environments and technological barriers have a great impact on the behavior of users, especially certain groups of people [9], while the supply chains' digitalization makes data-driven e-commerce solutions crucial for SMEs [10]. Thus, recent advancements suggest at least three gaps in the literature: inadequate causal interpretation of e-commerce diffusion drivers, limited consideration of spatial factors, and insufficient use of appropriate methodologies. This study aims to address all of them with the help of three methodologies: econometrics, machine learning, and network analysis. Econometrics allows making causal inference with account for endogeneity and unobserved heterogeneity; machine learning increases predictive power and reveals nonlinear relationships; clustering and network analysis help to find patterns and dependencies among different variables. The framework underlying this study relies on the combination of three theories. First, according to the TOE model, technology adoption is influenced by technological readiness, organizational skills, and environment factors. Second, new economic geography stresses the importance of the spatial dimension in economic activities and its implications, including infrastructural disparity. Third, institutional theory states that institutions can affect digital behavior and regional dynamics. As discussed above, such a framework allows making a multidimensional analysis of the e-commerce phenomenon that arises in the interaction of technological, socio-economic, and institutional elements. This combination also guarantees that chosen methods are consistent with the research objectives: econometrics will be used to make causal inferences; machine learning helps to understand the nature of observed phenomena; clustering and network analysis reveal group characteristics and structural features. By combining several methodologies and considering regional aspects, the present study makes three major contributions to the field: incorporation of causality into predictive models, explicit consideration of spatial dynamics, and longitudinal, region-level analysis of e-commerce diffusion in Italy. Finally, Italy appears to be a particularly suitable case for the current study because of its well-known North–South gap. While northern and central regions are characterized by developed infrastructure and a high level of innovation, their southern and insular counterparts experience digital convergence.

The article is structured as follows: Section 2 reviews key theoretical and empirical studies on regional digitalization, e-commerce diffusion, and technology adoption. Section 3 details data sources, variables, and the methodological framework, including econometric, clustering, and machine learning tools. Section 4 analyzes regional e-commerce trends in Italy (2013–2023) and highlights spatial inequalities. Section 5 presents

k-means clustering results on digital maturity, while Section 6 provides econometric estimates using IV and GMM models. Sections 7–9 discuss machine learning predictions, network analysis, and policy implications. Section 10 concludes with key findings and future research directions. Appendix A reports descriptive statistics.

2. Literature Review

In relation to the reviewed topics, the scholarly literature on e-commerce adoption in SMEs can be classified into several mutually connected categories: the impact of different determinants on adoption decisions and their outcomes; issues related to digital regional disparities; and sophisticated applications of quantitative methods for exploring heterogeneity. The first strand is dedicated to the analysis of technological and organizational determinants of e-commerce adoption and its impact on performance. Numerous studies confirm the important roles of entrepreneurial capabilities, human capital, and technological infrastructure in firms' success in digital transformation processes [11,12]. According to researchers, e-commerce serves as a mechanism that links firm capabilities to external opportunities [13], which are determined by the compatibility between external opportunities and internal capabilities [14]. Overall, the above-mentioned literature provides insights into the nature of adoption and performance, viewing them as multi-dimensional phenomena that result from the interaction between internal firm capabilities and external conditions. Another strand focuses on strategic and technological foundations of competitive e-commerce development. The papers analyzed in this context emphasize the importance of digital infrastructures, platform-oriented business models, and innovation capabilities in shaping e-commerce adoption and success [15–17]. Simultaneously, e-commerce adoption and development processes are increasingly viewed as integral components of broader economic processes of digital transformation, sustainability transition, and financial sector innovation [18]. The abovementioned studies imply that e-commerce adoption and development should be considered an integral part of firms' broader strategies. The third strand is centered on the discussion of contextual and institutional factors. In this context, recent research demonstrates that exogenous events, such as a pandemic, can accelerate adoption and expose structural problems in firms' technological capabilities [2,19]. From a broader perspective, scholars have paid considerable attention to the roles of infrastructure availability, legislation, digital literacy, and social capital in the successful diffusion of e-commerce [1,20–22]. Overall, all the abovementioned streams provide evidence for the impact of technological and institutional factors on SME e-commerce adoption and performance. However, there are several research lacunas, such as a lack of focus on regional aspects of adoption processes, an absence of integration among various approaches, and limited discussion of the interactions among technological, socio-economic, and institutional factors. The current study aims to bridge existing gaps in the literature on the adoption and diffusion of e-commerce technologies by applying an integrated approach combining analytical tools. In particular, it will employ an integrated approach that combines econometric techniques, machine learning methods, cluster analysis, and network analysis. See Table 1.

Table 1. Thematic overview of literature on e-commerce, innovation, and regional development.

Macro-Theme	Study (Author, Year)	Methodology	Key Findings
Tourism Attraction and Regionality	[5,6,23–32].	Case studies, spatial econometrics, regional comparative analysis, GIS mapping, survey-based tourism evaluation	These studies identify regional disparities in tourism attractiveness and highlight the role of cultural assets and local policies in shaping regional competitiveness, showing how tourism innovation contributes to economic growth.
E-Commerce and Digital Transformation	[3,4,17,33–42].	Econometric modeling, regression analysis, structural equation modeling (SEM), case studies, survey-based digital readiness assessment	These contributions demonstrate the positive impact of e-commerce adoption on SME performance and resilience, emphasizing digital transformation as a key driver of competitiveness while also identifying infrastructural and strategic barriers.
Technological Innovation and SMEs	[11,12,18,43–52].	Panel data analysis, innovation surveys, TOE framework, institutional theory, qualitative interviews	The literature confirms that technological innovation enhances SME productivity and competitiveness, with human capital and infrastructure acting as key mediators, and highlights the importance of innovation ecosystems and policy support.
Digital Literacy and Digital Divide	[53–61].	Descriptive statistics, sociological surveys, index construction, digital inclusion assessment, comparative regional studies	These studies reveal significant disparities in digital access and skills, showing that higher digital literacy fosters e-commerce adoption and innovation, and emphasizing education and infrastructure as key enablers.
Machine Learning and AI	[62–74].	Predictive modeling (SVM, Random Forest, Gradient Boosting), k-means clustering, supervised and unsupervised learning	The literature shows that machine learning models, particularly ensemble methods, achieve high predictive accuracy and are effective in identifying patterns in SME digital adoption and regional disparities.
COVID-19 and Economic Impacts	[1,2,19,75–80].	Comparative analysis, longitudinal studies, pandemic impact surveys, econometric resilience models	These studies highlight that e-commerce adoption mitigated the economic impact of COVID-19, enhanced SME resilience, and accelerated digital transformation, while also exposing structural weaknesses.

Note. This table categorizes key studies by theme, methodology, and core findings on digital and regional transformation.

3. Data and Methodology

In this section, the framework is operationalized via an empirical approach, elaborating on the data, variables, and the analytical approach adopted in the research. The choice of variables used in this study is influenced by both theoretical and empirical literature on

digital transformation, regional development, and the diffusion of e-commerce [81–83]. The dependent variable in this research is Ecombiz—the proportion of firms with at least ten employees engaging in online sales. This is a regional indicator of e-commerce diffusion. It measures the extent of digital business penetration, as well as more complex structures of competitive advantage, as indicated by the existing literature, which highlights the link between firms' digitalization and regions' socio-economic and infrastructural characteristics [83]. The independent variables used in the research cover different dimensions of digital transformation. In particular, regular internet users (Intuse) and household access to computers and the internet (Icthh) serve as measures of individual and household digital diffusion, respectively. Thus, Intuse measures the extent of digital participation among individuals, while Icthh measures the extent of digital readiness among households, indicating the level of technology accessibility required for participating in digital commerce. The above variables are considered endogenous as there exists a two-way relationship between them and the adoption of e-commerce [81,84]. Other control variables include Innovsys (innovation capacity of the production system), Gdpcpc (per capita gross disposable income), and Teleworking (Telework), which represent the foundations of technological, economic, and behavioral development at the regional level [82]. To address endogeneity and account for other contextual factors, the model incorporated instrumental and exogenous variables derived from ISTAT-BES indicator data. These variables represent different aspects of human, cultural, and social capital. First, Hs2564, Tq3034, and Contedu reflect residents' educational levels and lifelong learning activities [83]. Cultural practices and civic participation are represented by Readbooks and Civicpart, respectively. On the other hand, Nonprofit and Gentrust represent the institutional dimension in terms of social network structures and civic trust within the regions [84]. Environmental quality and broadband access in regions are represented by Urbangreen and Broadband, respectively. The combination of the above variables enables the model to consider multiple dimensions in explaining e-commerce adoption at the regional level. It can be noted that the above design follows the Technology–Organization–Environment (TOE) and Institutional theoretical frameworks [81,82,85]. Table 2 below provides an overview of the variables used in the empirical analysis, including their corresponding codes and names.

Empirically, the study relies on a multi-method approach, using each method to fulfill specific tasks, such as identifying causes, making predictions, and detecting patterns.

Firstly, the identification of causes will be conducted using econometric instruments for causal analysis, which account for endogeneity arising from reverse causality and omitted variables. In addition, it is assumed that time and space heterogeneity will need to be incorporated into this process [86,87].

Secondly, machine learning techniques will be employed to complement traditional econometric methods, providing an efficient way to capture non-linearities and complex interactions among variables. In particular, the application of Regularized Linear Regression, Random Forests, and Gradient Boosting is expected to improve model accuracy and feature importance assessment.

Thirdly, clustering analysis will be performed to assess regional heterogeneity. The use of the Silhouette coefficient-based K-means algorithm enables the identification of homogeneous regions with unique combinations of socio-economic characteristics, infrastructure conditions, and digital readiness [88].

Finally, network analysis will be applied to explore interdependencies between the technological, social, and economic variables. Using measures of centrality, including betweenness, closeness, and expected influence, it will be possible to determine which variables act as connectors within the digital ecosystem [89].

In general, this combination of empirical approaches constitutes an integrative multidimensional framework for studying e-commerce diffusion. See Figure 1.

Table 2. Variable description and classification.

Type	Variable (Full Name)	Code	Definition
Dependent	Share of firms with at least 10 employees engaged in online sales	Ecombiz	Businesses with at least 10 employees with online sales to end customers
Explanatory (Endogenous)	Regular internet users	Intuse	Individuals who use the internet frequently
	Household access to computers and internet	Icthh	Availability of at least one computer and an internet connection per household
Explanatory (Exogenous)	Innovation of the production system	Innovsys	Measure of innovation capacity in regional production systems
	Gross disposable income per capita	Gdpcpc	Per capita gross disposable income
Instrumental	Teleworking (remote work)	Telework	Employed people working from home
	People with at least a high school diploma (25–64 years)	Hs2564	Share of population aged 25–64 with at least upper secondary education
	Graduates and tertiary qualifications (30–34 years)	Tq3034	Share of population aged 30–34 with tertiary education
	Participation in continuing education	Contedu	Participation rate in lifelong learning
	Cultural participation outside the home	Cultout	Participation in cultural activities
	Reading books and newspapers	Readbooks	Frequency of reading books and newspapers
	Civic and political participation	Civicpart	Engagement in civic and political activities
	Non-profit organizations	Nonprofit	Density of non-profit organizations
	Generalized trust	Gentrust	Level of interpersonal trust
	Availability of urban greenery	Urbangreen	Access to green urban areas
	Ultra-fast fixed-line internet coverage	Broadband	Availability of high-speed broadband infrastructure

Source: ISTAT-BES: https://www.istat.it/statistiche-per-temi/focus/benessere-e-sostenibilita/la-misurazione-del-benessere-bes/gli-indicatori-del-bes/ (accessed on 12 December 2025).

Why Italy Matters: A Benchmark Case for Regional Inequality and Digital Transformation. The peculiarities of Italy's geography make it a special case study for investigating the relationships between digital transformation and regional disparities. Indeed, Italy is a highly territorially differentiated country, with three regions standing out: the advanced, technologically advanced North, the underdeveloped South, and the inland areas characterized by high structural heterogeneity. In this regard, Italy is particularly interesting for analyzing uneven trends in e-commerce adoption [90]. Moreover, such territorial differences encompass multiple dimensions beyond income levels, including gaps in infrastructure, innovation potential, and technological readiness [91]. This makes it possible to consider Italy a promising case for exploring complex processes involving interactions among various elements of the development environment. In this respect, it should be noted that digitalization, in general, and e-commerce in particular, can be considered a means of compensating for territorial disparities. Digital technologies enable firms to enter new markets and expand their business, regardless of location, promoting local entrepreneurship and greater market inclusion [92]. At the same time, given that Italy's productive structure consists mainly of SMEs, industrial districts, and production clusters, digital tools can positively influence competitiveness and innovation. In addition, e-commerce processes are linked to the availability of digital infrastructure and favorable policy conditions. Thanks to opportunities to overcome market-entry barriers offered by digital platforms, innovation and job creation can become more intensive, especially in lagging regions [93]. In its turn, such processes are encouraged by national and European policies aimed at supporting the development of these aspects, namely the PNRR [94]. In conclusion, Italy can be considered

a unique empirical and policy-oriented laboratory for studying the interaction between digital transformation and regional disparities.

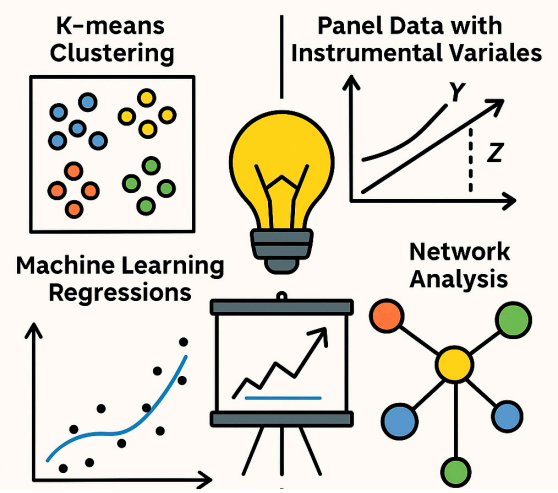


Figure 1. Integrated methodological framework for analyzing e-commerce adoption. Note: Visual representation of the four core methodologies used in the study: Instrumental Variable Panel Data, Machine Learning Regressions, K-Means Clustering, and Network Analysis. The central lightbulb symbolizes innovation in analytical integration.

4. Regional Dynamics of E-Commerce Adoption in Italy (2013–2023)

The current part presents the results of a descriptive analysis aimed at examining the dynamics of e-commerce implementation across different regions of Italy during the period 2013–2023. The focus is on firms with a staff of at least 10 employees performing transactions through e-commerce. For the purposes of the study, two key indicators have been used to estimate the extent of e-commerce adoption: absolute change in percentages during the time period studied and percentage change as a measure of relative growth of digital adoption. Thus, the use of both indicators allows for estimating the scale and intensity of e-commerce implementation. Based on the data obtained, it can be concluded that e-commerce development is increasing in Italy. However, it shows great variability across regions. The regions of northern Italy—Piemonte, Lombardia, Veneto, and Emilia-Romagna—are marked by consistent growth in the share of firms adopting e-commerce. For example, in Piemonte, the number of firms using e-commerce increased by 8.7 percentage points (241.67% growth). It means rapid adoption of digital processes, facilitated by infrastructure availability and adaptation capabilities. The relative increase in Lombardia was lower (5.5 percentage points, or 119.57%), indicating a more advanced economic setting in which some aspects of digital operations had already developed by the beginning of the period. See Figure 2.

The above regional patterns clearly demonstrate the importance of advanced manufacturing, proactive regional governance, and conducive local conditions for stimulating e-commerce development in Italy [95,96]. Among the most advanced regions is Trentino-Alto Adige, where the e-commerce level increased from 14.2% in 2013 to 25.8% in 2023 (+11.6 percentage points and an 81.69% relative increase), indicating maturity in the digital economy [97]. In second place was Friuli Venezia Giulia, where it increased by 7.8 points (+147.17%), while in Liguria it showed irregular but significant growth linked to port logistics and innovation among small businesses [98].The smallest region, Valle d'Aosta, saw e-commerce grow from 11% to 25.4% (+130.91%), demonstrating the flexibility of small, tourism-oriented economies in adopting new technologies [99]. In Central Italy, Tuscany, Lazio, Umbria, and Marche are experiencing significant growth in creative industries and services. Specifically, Toscana doubled its share to 13.8% [97], Lazio tripled it to

15.8%, being influenced by digitalization in Rome related to its ICT sector [100]; Umbria showed the highest national growth in e-commerce adoption by +410% due to local policy interventions [101]; and Marche grew by +162.79% due to SME innovations [102]. Southern Italy also showed impressive acceleration, with Campania and Puglia growing more than twice (+200%) as a result of investments in infrastructure and digital platforms [95,103]. Calabria grew by 20.3%, a 153.75% increase, demonstrating the potential for e-commerce diversification [102]. Sicily and Sardinia, with their focus on tourism and SMEs, recorded almost 3-fold increases (297.87% and 276.39%, respectively, with the latter showing the highest absolute growth in Italy). Meanwhile, Basilicata experienced a tremendous growth of 455% from 2% to 11.1%, demonstrating EU-financed innovation and entrepreneurship [97], while Molise showed 232% growth in the same period, proving that even small regions can overcome the digital divide [101]. Regionally averaged data show that, in terms of maturity, Trentino-Alto Adige (19.4%) and Valle d'Aosta (18.2%) lead the country; meanwhile, the south shows strong leaders, such as Sicily, Sardinia, and Calabria, each reaching about 12%. Lower average values in Molise, Marche, and Piemonte imply slower but consistent adoption. An important milestone occurred around 2020, as almost all regions experienced an acceleration of their digital transition due to COVID-19 [98]. Even though some regions experienced regression after the pandemic, the increased base remained stable, indicating that behavioral changes were sustainable. Altogether, the trajectory from 2013 to 2023 demonstrates that Italy has entered a period of digital maturity. Economically speaking, this finding indicates that the diffusion of e-commerce is increasingly becoming a factor of competitive advantage, helping companies (particularly in underdeveloped regions) overcome market boundaries and reach a wider audience. From this point of view, the importance of digital transformation extends beyond technology itself into a process capable of reducing inequalities. Although northern regions benefit from existing infrastructure, there is significant relative dynamism in the south and smaller regions, implying the convergence in the digital divide. E-commerce has become an integral part of competitiveness and regional development. Continued progress will require supporting policies that promote broadband development, innovation networks, and digital skills.

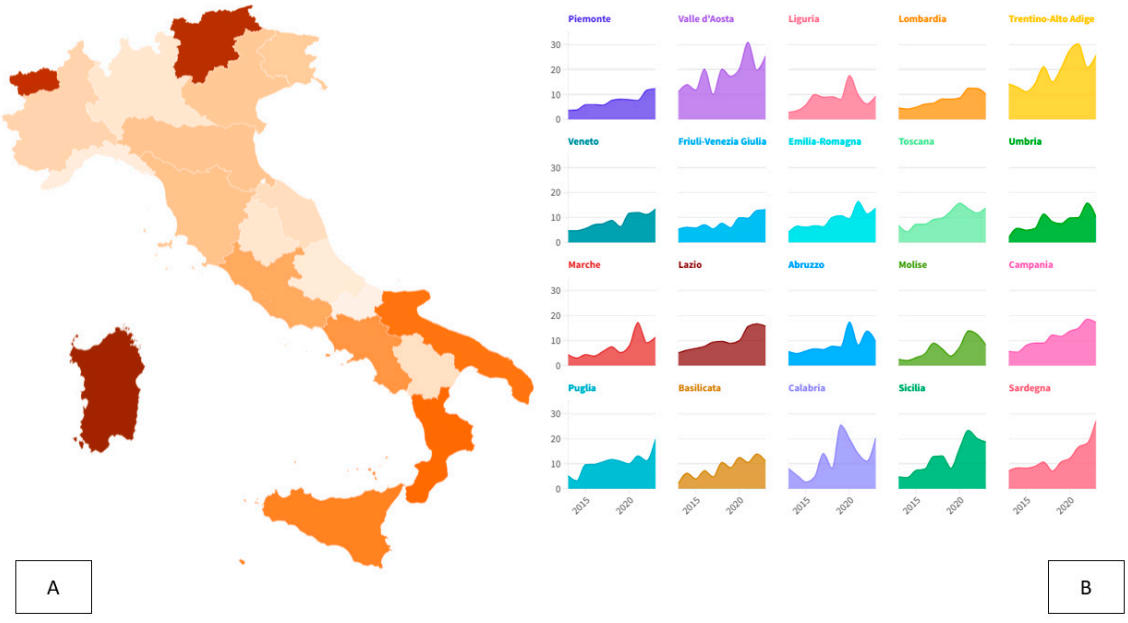


Figure 2. Regional patterns of e-commerce adoption among Italian businesses. Note: Part (**A**) shows the level of e-commerce adoption in 2023 by firms with at least ten employees across Italian regions. Part (**B**) illustrates the evolution of the same variable from 2013 to 2023 for each individual region, highlighting temporal and spatial disparities.

5. Digital Ecosystem Typologies in Italy: Insights from K-Means Clustering and Silhouette Analysis

However, while significant differences in e-commerce adoption across regions were found, consistent patterns of technological behavior across Italian regions have not been demonstrated. As such, this section aims to investigate the extent to which Italian regions can be grouped according to shared patterns of digital technology use.

K-means clustering optimization, based on the maximization of Silhouette coefficients, enables an interesting analysis of digital transformation across the twenty regions of Italy [73,104]. In the proposed model, the dependent variable is the share of firms adopting online sales strategies, while explanatory factors include individual internet diffusion (Intuse), household digital penetration (Icthh), innovation systems (Innovsys), per capita GDP (Gdpcpc), and teleworking (Telework). Collectively, these variables capture several crucial aspects of digital development, namely its technological and behavioral components at the household and firm levels [105,106]. Practically, it means that greater levels of internet diffusion and household connectivity correlate with e-commerce adoption since companies operate in an environment where consumers adopt modern digital technologies.

Notably, the larger the values of Intuse and Icthh, the higher the probability of classification as belonging to the group of regions with a more advanced digital ecosystem (Cluster C1), while lower levels of diffusion are typical of regions in Cluster C2, indicating lower levels of digital transformation. As Silhouette coefficient optimization maximizes within-group cohesion and group differentiation, the proposed model identifies two clusters of Italian regions. The first one includes the majority of the northern and central parts of the country—Piemonte, Valle d'Aosta, Liguria, Lombardia, Trentino-Alto Adige, Veneto, Friuli-Venezia Giulia, Emilia-Romagna, Toscana, Umbria, Marche, and Lazio. The second cluster comprises the southern regions—Abruzzo, Molise, Campania, Puglia, Basilicata, Calabria, Sicily, and Sardinia.

As already mentioned, this north–south divide reflects the known digital gap in Italy [107]. However, while emphasizing the differences between these regions, the clustering model also highlights the heterogeneity of digital-adaptation patterns driven by structural, economic, and behavioral changes. Clustering C1 is associated with the economically advanced regions featuring a well-developed infrastructure and innovative capabilities. Their Silhouette scores range from 0.52 to 0.68, indicating relatively good coherence. Notably, the top-performing regions are those that best embody the digital archetype—Emilia-Romagna (0.676), Friuli Venezia Giulia (0.674), and Lombardia (0.671) [108,109]. Other regions with high coherence and belonging to C1 are Trentino-Alto Adige (0.669) and Veneto (0.656). Their characteristics can be attributed to well-functioning small enterprises' ecosystems and efficient policies. Toscana (0.654), Piemonte (0.654), and Liguria (0.609) show lower coherence levels due to diversified economies, with creative (Toscana), manufacturing (Piemonte), and service-based sectors (Liguria). Meanwhile, Valle d'Aosta (0.527), Umbria (0.564), and Marche (0.575) occupy peripheral positions due to their economic structures. Finally, Lazio (0.630), whose major city is Rome, is characterized by a high level of urbanization and institutional power compensating for structural constraints, hence its place in the digital mainstream [110]. In contrast to clustering C1, the latter (C2) is associated with south-ern regions, featuring the latter, but fast digital advancement. Their Silhouette coefficients range from 0.53 to 0.69, indicating good coherence, especially for Basilicata (0.693), Calabria (0.691), Sicily (0.692), and Puglia (0.688). These regions are experiencing rapid growth driven by infrastructure expansion and digital entrepreneurship, alongside appropriate public policies. The high Silhouette score for Basilicata indicates convergence despite structural barriers. See Figure 3.

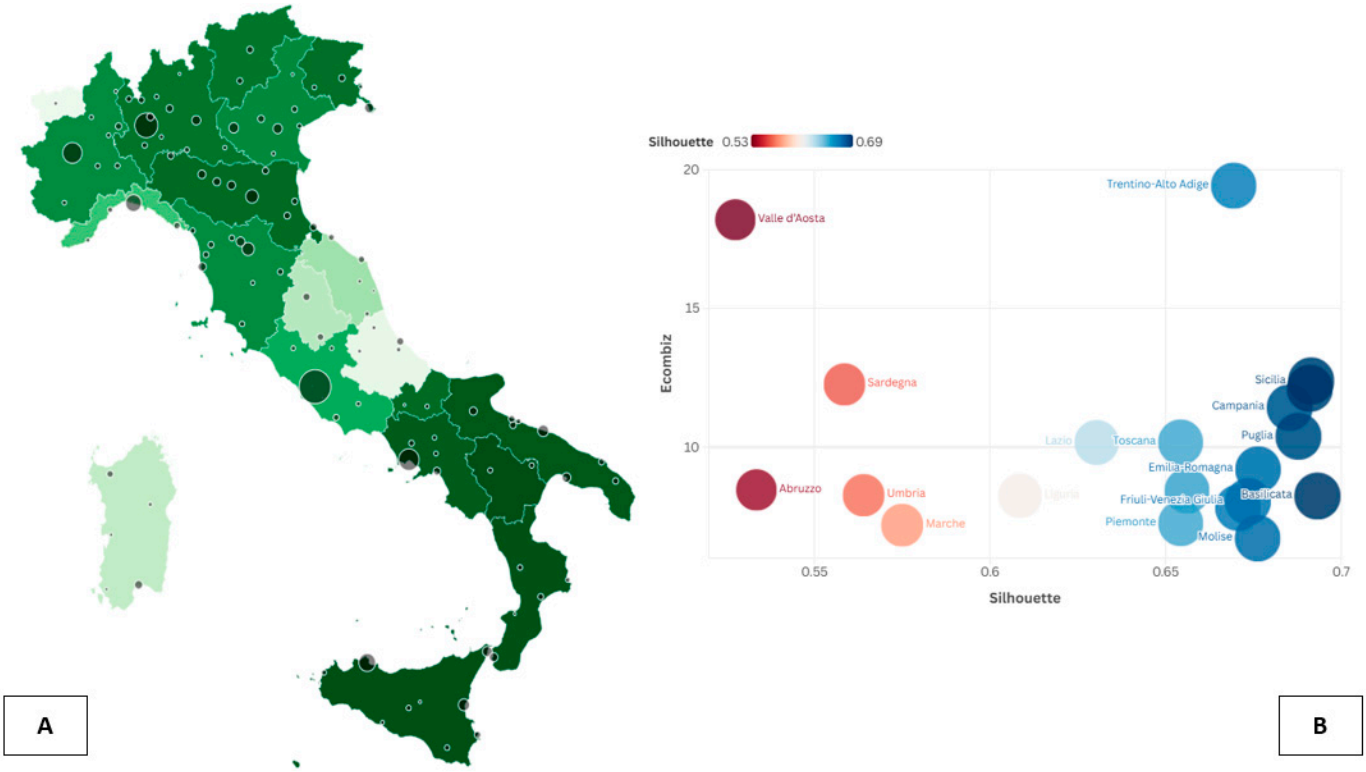


Figure 3. Regional distribution and clustering of e-commerce adoption in Italy. Note: Panel (**A**) displays the average value of e-commerce adoption by firms with at least ten employees across Italian regions from 2013 to 2023. Panel (**B**) visualizes the results of the k-means clustering using Silhouette Coefficients, identifying digital ecosystem patterns and cohesion levels among regions.

Next in line is Campania (0.685375) due to its rapid growth in the digital economy, associated with start-ups and a young, digitally literate population [111]. Both Calabria and Sicily show strong similarity to the cluster's silhouette profile (values above 0.69), underscoring the tendency of the southern regions to cluster together. E-commerce development in the south often occurs out of necessity, as digital technology offers new possibilities for businesses facing limited geographic and economic opportunities [112]. Puglia, finally, achieves the best result among the remaining regions, with a score of 0.68786, indicating its success in expanding infrastructure and digital engagement to become a leader in the south's ICT development.

Thus, based on the consistency of these results, one can state that the current digital gap between the north and the south of Italy no longer lies merely in technological disparities but is more closely related to socio-economic and institutional factors underlying digital routines and business models across different regions [113].

Moving further down the Silhouette coefficients, one can see that Abruzzo (0.533443) and Sardegna (0.558523) have relatively low cohesion levels. This can be attributed to their partially different structure compared to other central regions, including geographical remoteness and distance from both the center-north of Italy and its industry and services sectors. Moreover, the moderate level of digital engagement observed in Abruzzo can be better explained by its structural diversity than by its digital decline. Similarly, the relatively low Silhouette value of Sardinia, aside from higher e-commerce development, is due to the island character of the local economy, which has been accelerating its digitalization through tourism and logistics, yet remains less involved in industrial supply chains. On the contrary, Molise displays a fairly high Silhouette value (0.676238), indicating that, despite its smaller size, its economic structure aligns with the digital south development model.

Comparing the characteristics of the two clusters reveals certain trends. Cluster C1 shows a well-developed digitalization model characterized by a well-established ICT infrastructure, a relatively high level of income, and innovative capacity. Digital technology is deeply embedded in the local economy, and e-commerce development reflects an integral element of the digital ecosystem in terms of production dynamics, labor market, and consumer behavior. Local firms have integrated digital tools as integral elements of competitiveness rather than merely as adaptive mechanisms [111]. The other cluster demonstrates a recently developing model of digitalization with a rapidly growing yet relatively immature economic structure. Digitalization processes in this cluster are triggered by the need to adopt policies in response to COVID-19-related changes [112]. Thus, the high internal cohesion of Cluster C2 can be explained by the fact that the southern regions share similar digitalization trajectories and processes, with digitalization serving as a compensatory factor for their lagging history. Moreover, relatively high average Silhouette values indicate that the two clusters were effectively separated, with values in the 0.60–0.69 range and little overlap. Therefore, the separation of these clusters proves statistically and conceptually sound [73,106]. Nevertheless, some regions have medium Silhouette values, suggesting they may serve as intermediary cases where their regional digital systems converge.

As for the interaction between technological and other variables, Intuse and Icthh measure mostly technology-based aspects, namely access to and use of the internet and digital technology. At the same time, Gdpcpc, Innovsys, and Teleworking (Telework) measure other factors that shape the economic context within which the technology-based factors operate, thereby influencing firms' e-commerce performance.

From a policy point of view, this classification of the clusters allows for identifying areas of focus for stimulating digital economy development, in particular e-commerce. For Cluster C1 regions, maintaining and fostering innovation accumulation, along with integrating modern digital solutions into the production and distribution system, should be prioritized. For Cluster C2, the main focus should shift to consolidating recent achievements, promoting broadband internet accessibility, and improving digital capabilities among small companies and the rural population. The high level of cohesion in the cluster means that nationwide initiatives can positively impact the economies of many southern regions concurrently with the use of interregional collaboration programs.

To conclude, the k-means clustering analysis has demonstrated that clustering based on the Silhouette coefficient provides a clear picture of regional digitalization in Italy. Two identified clusters, C1 and C2, correspond to Italy's current state, characterized by two digital regions. The high level of cohesion within each cluster and the low silhouette coefficients for certain intermediary regions indicate their possible future roles. Thus, over the past 20 years, Italy has moved from a dispersed use of digital technology to a structured but heterogeneous ecosystem. See Figure 4.

The median values for Ecombiz among the clusters differ. The value of 8.37 was found for C1, which encompasses 12 regions located in northern and central Italy, whereas 10.93 can be noted for C2, which mostly includes regions located in the south and on islands of Italy. A discrepancy of 2.6 percent shows a higher level of use of e-commerce in C2 than in C1. At the same time, less variation in cluster C2 shows that this cluster is more homogeneous geographically. However, results must be analyzed with caution. Clustering highlights some patterns but does not allow establishing causality between the dependent variable and the factors causing it. In particular, clustering does not explain the size and even the sign of the effect technological, economic, and institutional factors have on the development of e-commerce. Therefore, econometric analysis is necessary.

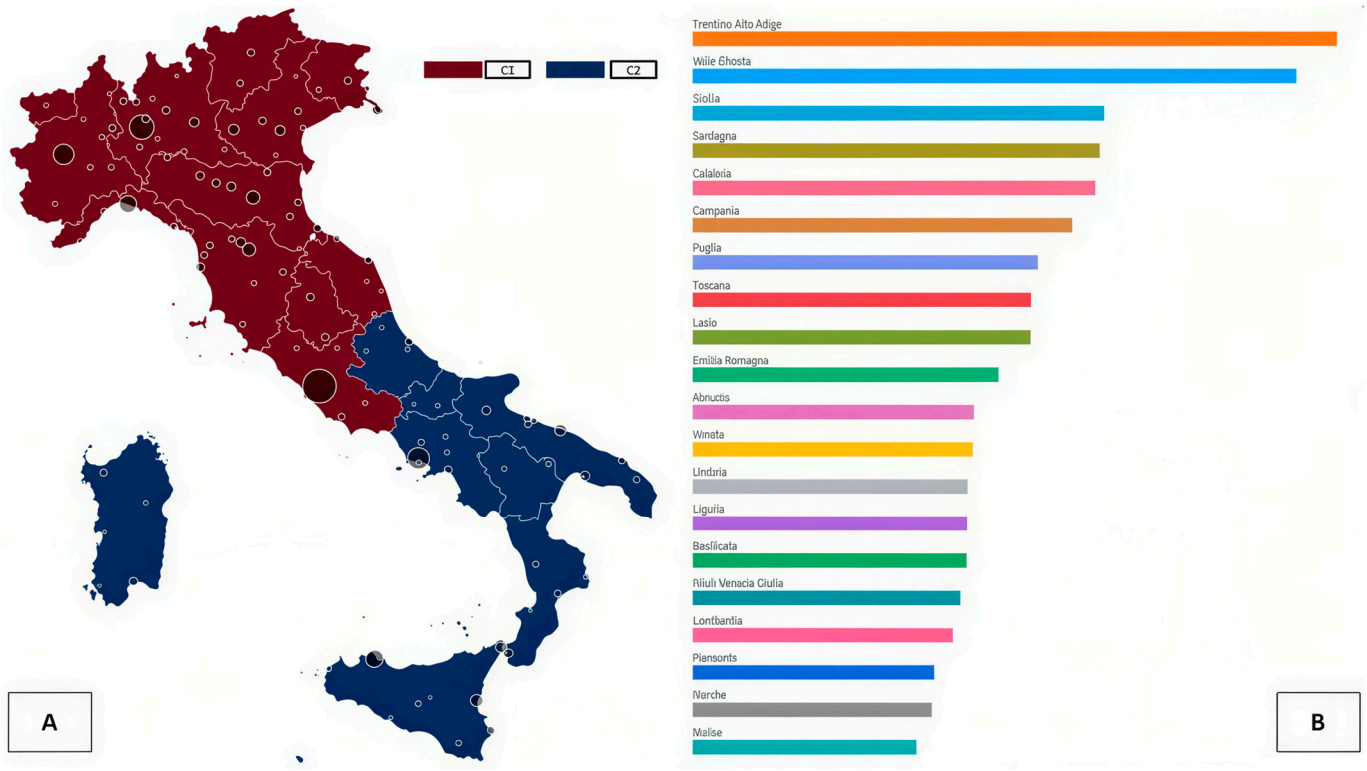


Figure 4. Regional clustering and average e-commerce adoption among Italian firms (2013–2023). Note: Panel (**A**) shows the spatial distribution of two clusters (C1 and C2) derived from k-means clustering of Italian regions based on e-commerce engagement and digital characteristics. Cluster C1 (in red) includes digitally mature regions, while Cluster C2 (in blue) comprises emerging digital regions. The size of the circles represents the population of the cities: the larger the city's population, the larger the circle. Panel (**B**) presents the average level of e-commerce adoption by firms with at least ten employees for each region during the period 2013–2023.

6. Econometric Determinants of Regional E-Commerce Adoption in Italy

Building on the identification of regional typologies based on descriptive analysis and clustering, as well as on digital typologies, the current section focuses on causal relationships. The focus of the paper will be the estimation of the direction and strength of the relationships between e-commerce diffusion and its determinants, while accounting for endogeneity. Previous sections provided evidence of regional variation in e-commerce diffusion; however, they did not attempt to identify its causes. To account for both the effects of various factors and endogeneity, econometric models will be used. Specifically, the current empirical model will analyze the structural determinants of e-commerce diffusion among firms in Italy that employ at least 10 workers and engage in online business. The main dependent variable is e-commerce diffusion (Ecombiz), which captures the degree of digital competitiveness. The explanatory variables are internet usage among the individuals (Intuse), ICT penetration in the households (Icthh), innovation capacity (Innovsys), per capita income (Gdpcpc), and the prevalence of teleworking (Telework). To address potential endogeneity in Intuse and Icthh, an instrumental variables (IV) approach will be used. The instrument set (Z) includes aspects of regional development associated with education, culture and civic engagement, social capital and trust, and digital infrastructure. The education component consists of Hs2564, Tq3034, and Contedu; the cultural and civic components consist of Cultout, Readbooks, and Civicpart; the social capital and trust components consist of Nonprofit and Gentrust; lastly, broadband internet and urban green spaces (Broadband and Urbangreen) are considered to be the elements of digital infrastructure. Together, these variables affect e-commerce diffusion by influencing human

capital formation, social capital, and the availability of digital resources. This approach provides the basis for analyzing the determinants of e-commerce diffusion by accounting for digital inclusion and innovation capacity, as well as the general socio-economic environment. For the specification of the empirical model, a fixed-effects IV (2SLS) approach will be used, which is appropriate for this study given its focus on endogeneity and regional heterogeneity.

Specifically, we have estimated the following equation:

$$Ecombiz_{it} = \alpha + \beta_1(Intuse)_{it} + \beta_2(Icthh)_{it} + \beta_3(Innovsys)_{it} + \beta_4(Gpdomcpc)_{it} + \beta_5(Telework)_{it} + \mu_i + \varepsilon_{it}$$

$$Z = [hs2564,\ Tq3034,\ Contedu,\ Cultout,\ Readbooks,\ Civicpart,\ Nonprofit,\ Gentrust,\ Urbangreen,\ Broadband]$$

$$i = 20\ t = [2004; 2023].$$

Econometric analysis explores the drivers of regional digitalization in Italy. The dependent variable (Ecombiz) measures the share of firms with at least 10 employees that conduct online sales within a specific region. The variable captures cross-regional variation in firms' e-commerce involvement—a defining aspect of digital development in Italy [114]. Several methods are employed to address endogeneity and unobservable heterogeneity: fixed-effect IV (2SLS), random-effect 2SLS, two-step GMM, and dynamic system GMM [115]. These methods differ in how they identify assumptions and in the type of analysis. For example, the fixed-effect IV approach accounts for unobservable, time-invariant variables and can be interpreted causally, whereas the random-effect method yields less biased regression coefficients when heterogeneity is uncorrelated with the regressors. The dynamic system GMM extends the scope of analysis by introducing lagged variables, thereby allowing the study of persistence and adjustment dynamics while addressing endogeneity via internal instruments. In general, the explanatory variables refer to the digital and economic structures of the region. The individual internet use indicator (Intuse) and the household penetration of ICTs (Icthh) are considered endogenous as the process of e-commerce diffusion is likely to be correlated with these two variables. To solve the problem of endogeneity, an instrumental variable technique is applied based on several proxies of education, culture, social activities, social capital, and infrastructure including Hs2564, Tq3034, Contedu, Cultout, Readbooks, Civicpart, Nonprofit, Gentrust, Urbangreen, and Broadband. These variables are expected to have an indirect impact on e-commerce diffusion through shaping human capital, social cohesiveness, and digital resource availability while being unrelated to the decision-making process among firms regarding e-commerce [52]. According to the fixed-effects IV (2SLS) results, Intuse was found to exert a significantly positive effect on e-commerce diffusion, with coefficients ranging from 0.73 to 0.85 (z = 6.3–14.0). Thus, an increase in internet penetration rates by one percentage point corresponds to an increase in the share of firms engaging in e-commerce by 0.7–0.8 percentage points, which proves the structural character of Internet penetration in digital diffusion. On the contrary, Icthh had a negative statistically significant association with e-commerce diffusion (-0.60 to -0.72, $p < 0.01$), which could be explained by saturation effects or a shift toward external digital platforms for e-commerce in developed regions. Finally, some variables have only a moderate impact on Ecombiz. In particular, Innovsys shows a negative association with e-commerce diffusion, suggesting that innovations in traditional manufacturing and production systems are not necessarily conducive to digital adoption. The Gdpcpc variable is positively associated with e-commerce, but only moderately so. As for telework, it shows a positive coefficient (approximately 0.11). It appears the variable reflects only the short-run impact of the COVID-19 period. The model diagnostics showed that the fixed-effects specification had an R^2 of 0.74, while IV and GMM estimations were validated for the absence of autocorrelation and for instrument relevance. Thus, it appears that digital

adoption in Italy is mainly driven by Internet penetration (introduced as a proxy for digital participation). On the other hand, socio-cultural factors such as educational attainment and civic participation also matter [52]. As follows, Italian policy should focus on enhancing the socio-economic and cultural context for digital diffusion, along with investing in digital infrastructure. For instance, in addition to infrastructure investments, policymakers need to promote digital capabilities and social cohesiveness. In conclusion, the econometric results provide empirical support for the theoretical model discussed above. Indeed, variables associated with the digital infrastructures and Internet usage are found to be important drivers of e-commerce diffusion, in line with the theoretical TOE framework. In addition, economic factors are insufficient to explain differences across Italian regions. See Table 3.

Table 3. Estimates from econometric models of e-commerce adoption.

Models	Variables	Coefficients	Std. Errors	z/t	$p > \|z\|$	[95% Conf. Interval]
Fixed-effects (within) IV—2SLS	Intuse	0.7551	0.0817	9.24	0.000	[0.5950; 0.9152]
	Icthh	−0.6191	0.0716	−8.64	0.000	[−0.7595; −0.4786]
	Innovsys	−0.0172	0.0056	−3.05	0.002	[−0.0283; −0.0061]
	Gdpcpc	0.000168	0.000052	3.21	0.001	[0.000065; 0.000270]
	Teleworking (Telework)	0.1201	0.0550	2.18	0.029	[0.0123; 0.2278]
	_cons	−20.903	11.232	−1.86	0.063	[−4.2917; 0.1111]
Fixed-effects—2-Step GMM	Intuse	0.7312	0.0522	14.00	0.000	[0.6288; 0.8336]
	Icthh	−0.6071	0.0475	−12.79	0.000	[−0.7001; −0.5140]
	Innovsys	−0.0141	0.0036	−3.86	0.000	[−0.0212; −0.0069]
	Gdpcpc	0.000166	0.000041	4.10	0.000	[0.000087; 0.000245]
	Teleworking (Telework)	0.1117	0.0313	3.57	0.000	[0.0503; 0.1731]
G2SLS Random-Effects IV	Intuse	0.8454	0.1069	7.91	0.000	[0.6360; 1.0548]
	Icthh	−0.7165	0.0979	−7.32	0.000	[−0.9083; −0.5246]
	Innovsys	−0.0191	0.0060	−3.20	0.001	[−0.0307; −0.0074]
	Gdpcpc	0.000144	0.000057	2.54	0.011	[0.000033; 0.000255]
	Teleworking (Telework)	0.0446	0.0655	0.68	0.496	[−0.0837; 0.1729]
	_cons	−0.8035	10.295	−0.78	0.435	[−2.8213; 1.2144]
Dynamic Panel Two-Step System GMM	Intuse	0.7566	0.1204	6.28	0.000	[0.5046; 1.0086]
	Icthh	−0.6442	0.1207	−5.34	0.000	[−0.8969; −0.3915]
	Innovsys	0.0081	0.0092	0.87	0.394	[−0.0113; 0.0274]
	Gdpcpc	0.000122	0.000071	1.72	0.102	[−0.000027; 0.000272]
	Teleworking (Telework)	0.0408	0.0929	0.44	0.666	[−0.1537; 0.2353]
	_cons	−0.8374	10.490	−0.80	0.435	[−3.0330; 1.3581]
Instrumental-Variables 2SLS Regression	Intuse	0.7551	0.0791	9.55	0.000	[0.6001; 0.9101]
	Icthh	−0.6191	0.0694	−8.92	0.000	[−0.7550; −0.4831]
	Innovsys	−0.0172	0.0055	−3.15	0.002	[−0.0279; −0.0065]
	Gdpcpc	0.000168	0.000051	3.32	0.001	[0.000069; 0.000267]
	Teleworking (Telework)	0.1201	0.0532	2.26	0.024	[0.0157; 0.2244]
	_cons	−51.715	12.768	−4.05	0.000	[−7.6741; −2.6690]

Note: This table reports coefficient estimates across five models examining determinants of regional e-commerce adoption.

This paper analyzes the determinants of business digitalization in Italian regions. The dependent variable is Ecombiz—firms' participation in online sales to end customers. This indicator describes firms' involvement in e-commerce, a core part of Italy's digital transition [52]. Several econometric models were used to analyze digi-tal infrastructure, innovative processes, and socio-economic determinants of the regional formation of e-commerce activity. Multiple procedures (Fixed-Effects IV (2SLS), G2SLS Random-Effects IV, Instrumental-Variables 2SLS, and System GMM (Two-Step)) provide robust analyses that can be considered to establish a cause-and-effect relation [114]. The explanatory variables are Intuse (regular internet users) and Icthh (computer and household internet penetration). Innovative production systems (Innovsys) reflect firms' technological competencies and flexibility, while per capita gross disposable income (Gdpcpc) indicates consumers' purchasing power. Telework measures changes in employment practices driven by digitalization. Instrumental variables (Z) represent schooling, sociocultural, and infrastructural characteristics of regions that affect the development of e-commerce activity. Variables Hs2564 and Tq3034 reflect the level of schooling, while continuing education (Contedu) is an indicator of workers' adaptability. The indicators readbooks, Civicpart (civic organization participation), and nonprofit (civil society organizations density) measure participation in the collection process and in the acceptance of innovations. Generalized trust (gentrust) is another indicator of innovation acceptance, while urbangreen and broadband describe environmental quality and digital infrastructure, correspondingly [96,97]. The Fixed-Effects IV (2SLS) model with 380 observations and 20 Italian regions explains approximately 74% of the regional variance in e-commerce activity, based on the within-R^2 of 0.7387. The value of $R^2 = 0.6559$ proves sufficient explanatory power of the model, while the value of Wald $\chi^2 = 500.94$ ($p < 0.001$) means that the explanatory power of the model is very high. Since the correlation between unobserved effects and regressors is negative (corr(u_i, Xb) = −0.1687), fixed effects are suitable, while the value of the intraclass correlation coefficient ($\rho = 0.3082$) confirms high regional heterogeneity. The G2SLS Random-Effects IV model also achieves high explanatory power, with identical within-model $R^2 = 0.7270$, total $R^2 = 0.6489$, and Wald $\chi^2 = 284.74$ ($p < 0.001$). Despite the low $\rho = 0.1766$, which reduces certainty about individual groups, the model provides advantages under the existing correlation structure. The explanatory power ($R^2 = 0.7580$) of the Instrumental-Variables (2SLS) cross-sectional model demonstrates its ability to make the best predictions. It is indicated by the Root MSE = 3.0341 of the model and significant values of F-statistic ($F(5, 19) = 56.13$) and Wald $\chi^2 = 31{,}394.17$ ($p < 0.001$) that prove the instrument quality and legitimacy [115]. Dynamic effects can be estimated using the System GMM (Two-Step) specification. The validity of this model is confirmed by Arellano-Bond AR(1) $z = -2.70$ ($p = 0.007$) and AR(2) $z = -0.01$ ($p = 0.994$). Moreover, the Hansen test ($\chi^2(9) = 9.82$, $p = 0.365$) and Diff-in-Hansen statistics confirm instrument exogeneity. All tests show that the digitalization process, as measured by the indicator Intuse, significantly enhances the diffusion of e-commerce activity across Italian regions, while other variables, such as Icthh, Innovsys, Gdpcpc, and Telework, also contribute to this effect. Education, lifelong learning, civic activities, and social capital are essential drivers of digitalization in Italian regions. Very high values of within-R^2 indicate that dynamics over time, rather than differences between Italian regions, play the major role in the process of diffusion of e-commerce activity.

To sum up, e-commerce development in Italy is a multivariate process driven by digital infrastructure, education, innovation, and social capital. While this paper provides robust evidence on the determinants of e-commerce diffusion, the generalizability of this evidence warrants further consideration, given its application to developed economies with considerable regional heterogeneity, an established institutional framework, and digital

infrastructure characteristic of Italy. In less digitally advanced countries or those with lower institutional capacity, the relative importance of socio-cultural and technological factors can differ. In addition, the analysis is conducted in a single-country context, which limits its external validity regarding economic institutions and the economy's structure. Future research will test this framework in cross-country analysis.

Overall, the findings indicate the robust and dynamic nature of digital transformation. Policymakers should consider stimulating broadband, improving workers' digital competence, promoting innovation, and enhancing social capital to create a more dynamic and efficient economy. Econometric models provide solid evidence about the causal relations between determinants and the outcome variable, but they do not completely describe all possible nonlinear relationships and predictions. Thus, the following section provides a complementary machine learning analysis.

To conclude, these findings should be used with care in developing countries or under other institutional circumstances. See Table 4.

Table 4. Model diagnostics and summary statistics for e-commerce adoption regressions.

Statistics	Fixed-Effects IV (2SLS)	G2SLS Random-Effects IV	Instrumental-Variables 2SLS	System GMM (Two-Step)
Number of Observations	380	380	380	380
Number of Gropus (n)	20	20	-	20
R-squared (within)	0.7387	0.7270	-	-
R-squared (between)	0.0419	0.0480	-	-
R-squared (overall)	0.6559	0.6489	0.7580	-
Wald χ^2	500.94	284.74	31,394.17	-
Prob > χ^2/F	0.0000	0.0000	0.0000	0.000
corr(u_i, Xb)	−0.1687	0 (assumed)	-	-
σ_u (var. individual)	2.0951	1.4537	-	-
σ_e (errore)	3.1392	3.1392	-	-
ρ (rho)	0.3082	0.1766	-	-
Root MSE	-	-	3.0341	-
Instruments	-	-	-	15
F(5, 19)	-	-	-	56.13
Arellano-Bond test AR(1)	-	-	-	z = −2.70, Pr > z = 0.007
Arellano-Bond test AR(2)	-	-	-	z = −0.01, Pr > z = 0.994
Sargan test	-	-	-	$\chi^2(9)$ = 38.38, Prob > χ^2 = 0.000
Hansen test	-	-	-	$\chi^2(9)$ = 9.82, Prob > χ^2 = 0.365
Diff-in-Hansen (GMM levels)	-	-	-	$\chi^2(2)$ = 0.49, Prob > χ^2 = 0.784
Diff-in-Hansen (GMM collapsed)	-	-	-	$\chi^2(4)$ = 2.10, Prob > χ^2 = 0.718

Note: This table presents key statistical indicators for evaluating model fit, consistency, and instrument validity across four econometric specifications used to estimate regional e-commerce adoption drivers.

7. Performance Evaluation of Machine Learning Algorithms for Predicting E-Commerce Adoption

In assessing algorithmic performance, it is essential to consider that some indicators are minimized while others are maximized. Here, the appropriate metrics—MSE, scaled MSE, RMSE, and MAE—constitute the error-based indicators, with lower values indicating superior performance. An equal comparison, in turn, entails interpreting smaller numbers as indicative of superior predictive accuracy and model efficacy [116,117]. Looking at the normalized outcomes, the artificial neural network (ANN) clearly records the highest error rates across all indicators, with all normalized values equaling 1.000, indicating poor performance among the considered models. In sharp contrast, models like the Regularized Linear, Random Forest, and KNN recurrently achieve the lowest minimized error scores,

indicative of superior ability to curb prediction errors and maintain model stability [118,119]. The regularized Linear model, in particular, shows superior consistency, whereby the recorded low levels of MSE (0.180), scaled MSE (0.062), RMSE (0.234), and MAE (0.068) indicate superior balancing of the trade-off of the balancing variance as well as an efficient ability in generalizing among the observations. The Random Forest model runs in parallel, and the similar low levels of committing errors indicate good adaptability to non-linearity, though the RMSE and MSE values are relatively higher. The KNN model also produces competitive results, especially in terms of MAE, yet it is vulnerable to data dimensionality and prone to local irregularities. Of all the models, the Regularized Linear algorithm is the best-balanced and most reliable. The overall reduction in errors across the metrics, coupled with computational efficiency and interpretability, makes the algorithm the most appropriate choice when both accuracy and model complexity are needed. In sharp contrast, ANN shows overfitting or convergence, while Decision Tree and Boosting-based models show fair, though stable, outcomes. Therefore, the Regularized Linear technique constitutes the superior compromise among accuracy, robustness, and generalizability (Figure 5).



Figure 5. Performance metrics across machine learning algorithms for predicting e-commerce adoption. Note: The figure compares seven machine learning algorithms—Boosting, Decision Tree, KNN, ANN, Random Forest, Regularized Linear, and SVM—based on four evaluation metrics: Mean Squared Error (MSE), Scaled MSE, Root Mean Squared Error (RMSE), and Mean Absolute Error to Median Absolute Deviation ratio (MAE/MAD). Lower values indicate better predictive performance. Boosting and Regularized Linear models show the best overall accuracy.

According to the Regularized Linear Regression results, the relative influence of predictors can be seen in the explanation of the share of firms that operate through online sales. Regarding mean dropout loss, Intuse (regular internet use) ranks first with an impact of 14.878, suggesting that internet penetration is crucial to e-commerce adoption [120]. The second place in terms of influence goes to Icthh (availability of ICTs in households), with a

mean dropout loss of 9.920, indicating that household-level access to the digital network is important [121]. As for other factors, their importance is lower. Thus, telework, with a mean dropout loss of 3.760, does not strongly affect the outcome [122]. In addition, Innovsys and Gdpcpc also have mean dropout loss values of 3.538 and 3.537, respectively, which imply that they play an enabling role rather than a structural one. Therefore, it should be stated that the findings of the study prove that digital connectivity and internet use are crucial factors that determine whether a certain market will adopt e-commerce practices or not. As for policies, the implications include increasing access to digital infrastructure in developing countries. See Figure 6.

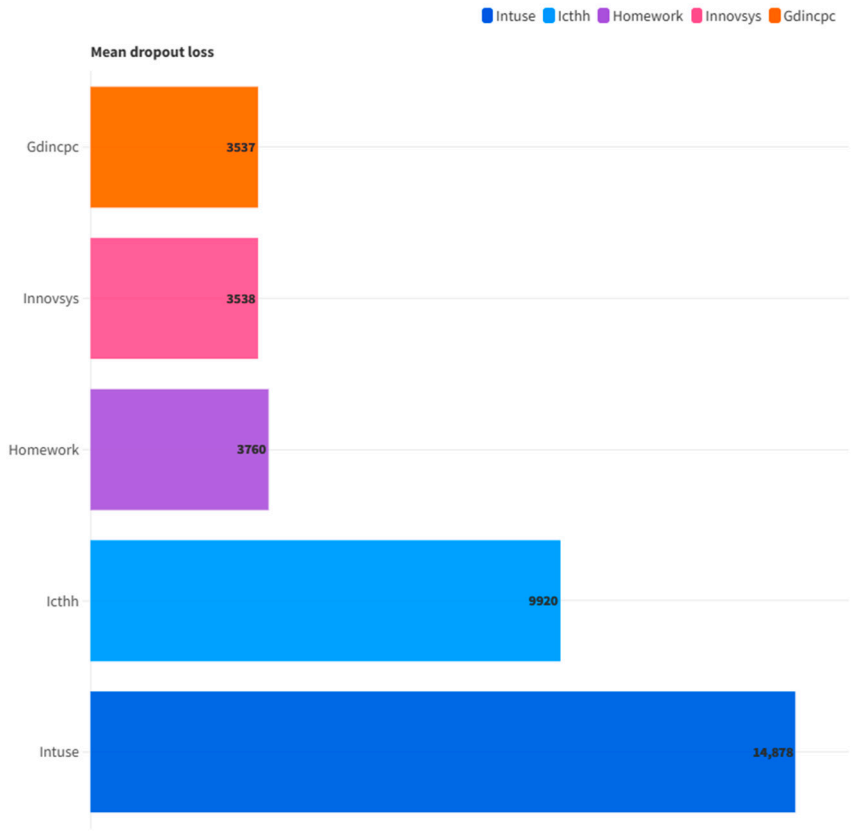


Figure 6. Variable importance in predicting e-commerce adoption using dropout loss (Neural Network Model). Note: This chart illustrates the mean dropout loss for each variable, indicating their relative importance in predicting the level of e-commerce adoption among firms. Higher dropout loss suggests a greater impact on model performance when that variable is removed, with *Intuse* (regular internet use) and *Icthh* (household digital access) being the most influential predictors.

The output generated by the Regularized Linear Regression algorithm, combined with an additive explanation of the forecasts for the test set, provides insight into the individual contributions of each variable to the model output. In this context, it is important to highlight the base value—5.177—which is the same across all four cases as the model's prediction in the absence of any descriptive variables [123]. Changes from the base value can be seen in how different features affect the output, either positively or negatively, to demonstrate how socioeconomic and technological factors affect companies' decisions to sell products online.

Among the analyzed variables, the Intuse variable, which represents the percentage of regular internet users, shows the greatest influence, with the highest negative impact (−26.740) in Case 4 and the highest positive impact (9.575) in Case 3. Such changes in the level of the effect indicate that the level of internet diffusion plays a crucial role in the formation of non-linear willingness to develop e-commerce [122,124].

Another important factor affecting the willingness to engage in e-commerce is Icthh—the number of households with access to a computer and the internet, whose impact is predominantly positive (21.048 in Case 4, 8.973 in Case 5). As shown, leveraging IT infrastructure enables companies to conduct online activities more efficiently [121]. Innovsys and Gdpcpc (production system innovation and gross disposable income per capita) have relatively low impacts with values close to zero. Consequently, innovation and income levels are not among the leading factors influencing e-commerce willingness [125].

Finally, the Telework variable—which indicates the level of teleworking in the region under consideration—affects the output relatively weakly and negatively. This suggests that the role of remote work is ambiguous and cannot be considered a driving force behind businesses' engagement in digital trade. Overall, internet diffusion and connectivity are among the major drivers of online commerce (Figure 7).

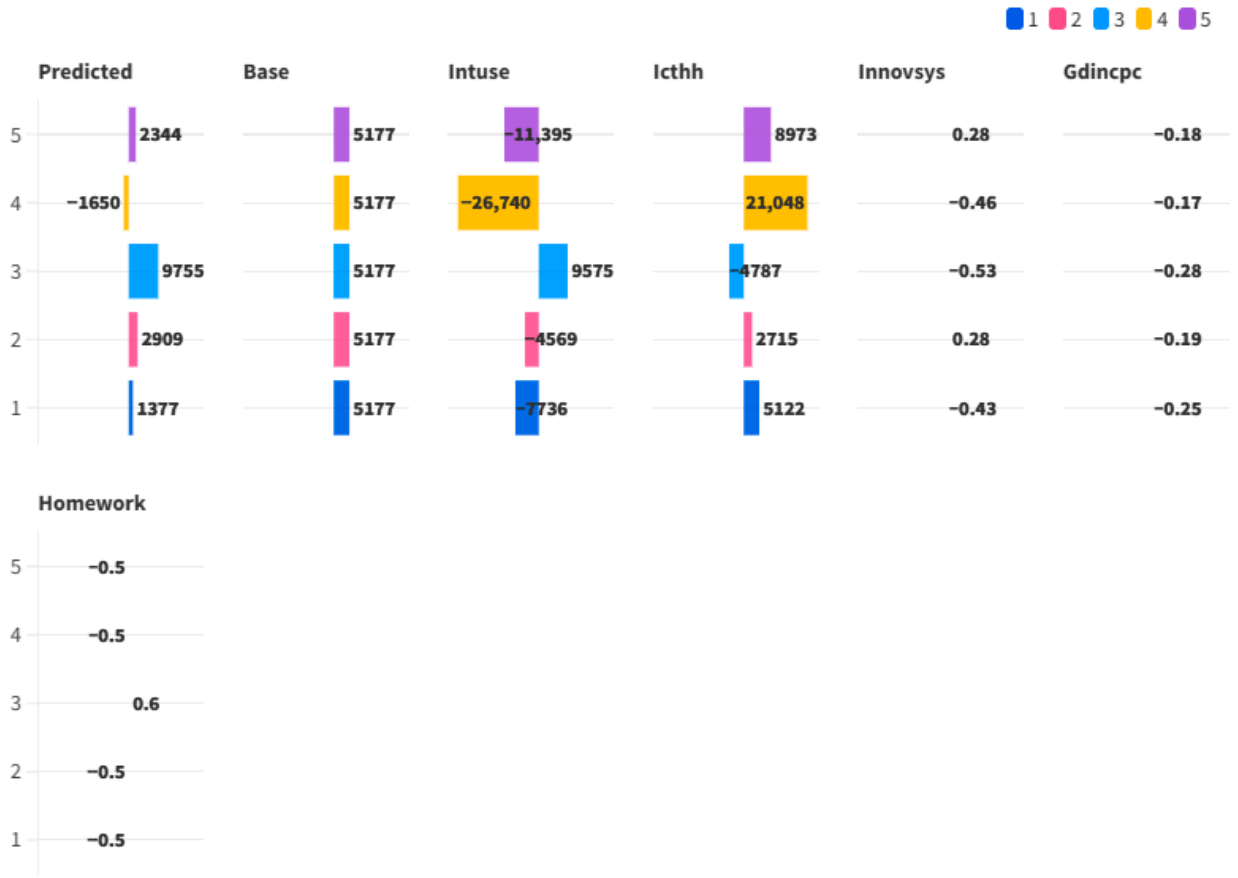


Figure 7. Additive decomposition of predictions from the regularized linear regression model. Note: Bar segments show each variable's contribution to predicted e-commerce adoption across five representative test cases.

The outcomes achieved by implementing the Regularized Linear Regression algorithm provide an explanation for the association between the chosen set of explanatory variables and the proportion of companies that employ at least 10 workers and sell their products online to consumers. The intercept is 5.078 and represents the baseline value of e-commerce activities when all predictors are considered simultaneously. The independent variable Intuse, measuring the proportion of regular internet users, has the highest positive coefficient ($\beta = 10.493$). In turn, such findings underscore the importance of internet penetration as a primary determinant that encourages companies to use e-commerce markets. Therefore, the higher the penetration rate, the greater the likelihood that a company will sell its goods and services online [126].

In addition, one independent variable has a negative coefficient (−6.834). It relates to Icthh and means having access to a household computer with an internet connection in at least one household within an area where the company operates. Such a coefficient can be explained by the substitution effect, which leads to online activities being associated more with consumer expenses than with business operations [127]. Furthermore, Innovsys and Gdpcpc have relatively insignificant negative coefficients (−0.343 and −0.311, respectively). These values indicate that neither innovation intensity nor income per capita has a significant impact on e-commerce volume growth [128].

In conclusion, it should be noted that another predictor variable has a positive coefficient (Telework = 0.937). This means that teleworking might be positively correlated with online sales and encourage companies to adopt digital technologies, as online work becomes increasingly popular [129]. Overall, it can be stated that internet use among the general population is one of the most influential factors contributing to e-commerce diffusion [123]. See Figure 8.

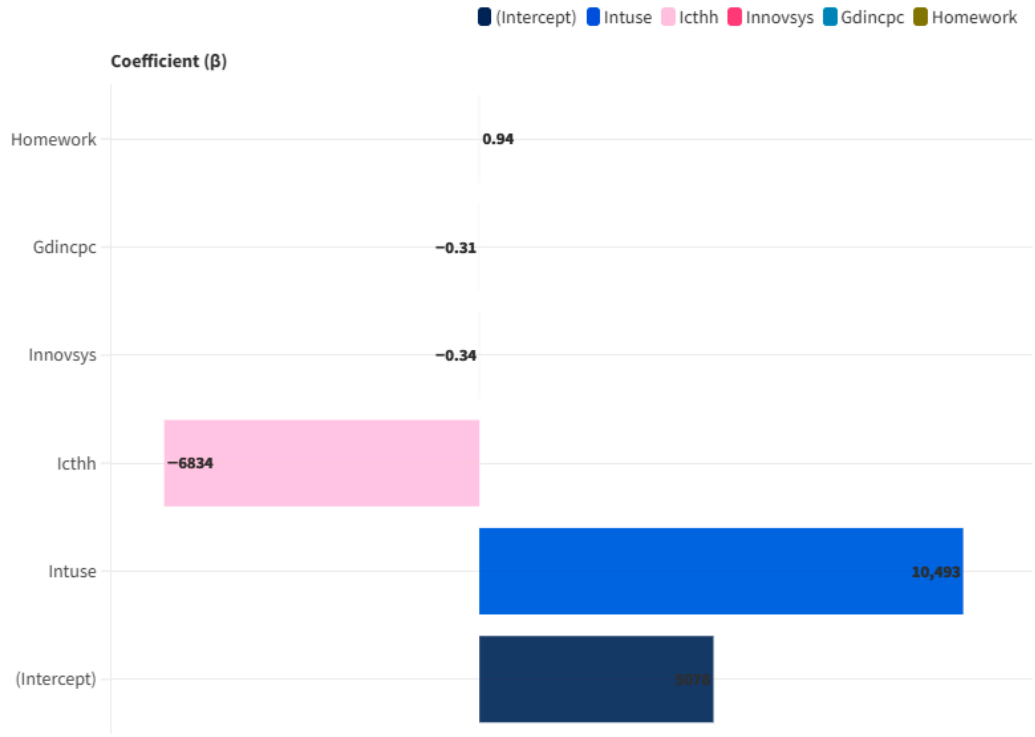


Figure 8. Coefficients from regularized linear regression model predicting regional e-commerce adoption. Note: Internet usage drives e-commerce adoption; other socioeconomic factors show weaker or inverse effects across regions.

The first panel illustrates the relationship between observed and predicted values of the test data. The results clearly indicate that the two are highly correlated, with the dots forming a cluster around the 45° line. This means that the Regularized Linear Regression Model achieved high accuracy and identified all factors affecting the proportion of businesses adopting online sales. Panel B provides a visual representation of how predictor coefficient paths change as the value of the regularization parameter (λ) increases. With an increase in λ, all coefficients tend to zero at different speeds. Here again, regularization is seen as a method for penalizing the model, reducing its complexity, and improving its generalization. Notably, the Intuse factor had the highest coefficient values consistently among the others. On the other hand, household connectivity had a lower impact than internet usage since Icthh dropped sharply. Furthermore, Innovsys, Gdpcpc, and Teleworking reached zero very quickly. Panel C shows the cross-validated mean squared error as a

function of λ. It is worth noting that the plot shows the familiar shape of the MSE curve, which is indicative of the bias-variance trade-off. Initially, as λ increases, the MSE decreases, but eventually it begins to rise. Thus, there are two dashed lines showing two particular values of λ. While the left dashed line represents the value minimizing the cross-validated MSE, the right dashed line represents the choice based on the so-called one-standard error rule (Figure 9).

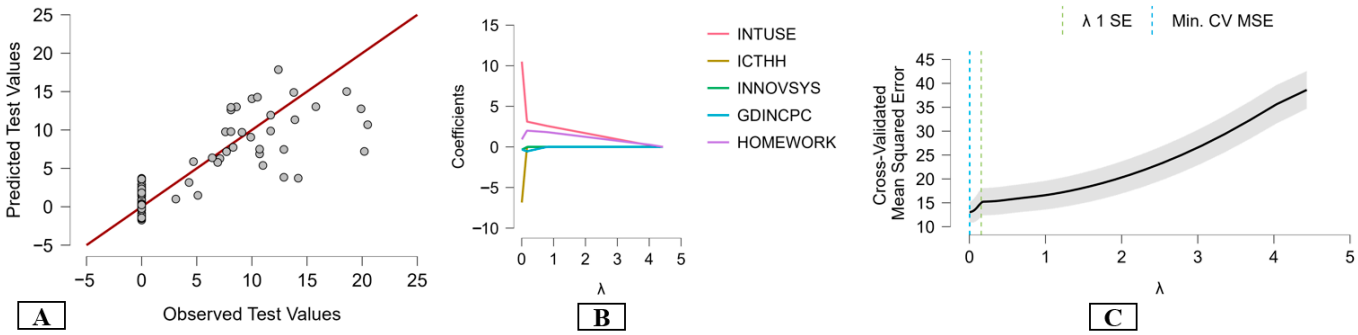


Figure 9. Performance and behavior of the regularized linear regression model. Note. Panel (**A**) shows the relationship between observed and predicted test values, indicating strong model accuracy. Panel (**B**) illustrates the shrinkage of predictor coefficients as the regularization parameter (λ) increases. Panel (**C**) reports the cross-validated mean squared error (MSE) as a function of λ; the dashed lines denote the λ value minimizing the MSE and the λ selected according to the one-standard-error rule.

8. Network Analysis of Digital and Socioeconomic Drivers of E-Commerce Adoption in Italy

The centrality analysis allows for the identification of the specific structure of relationships among the variables that affect companies' engagement in e-sales. Thus, it can be seen that Intuse (regular internet use) appears as the node with the highest values, particularly for indicators such as strength (1.200) and expected influence (1.955), which highlights the importance of the variable. Accordingly, this variable can be considered central, indicating the strong impact of internet use on other factors and serving as a driving force for e-commerce diffusion [130,131]. Moreover, the relatively high closeness value (0.552) indicates strong connections with other nodes, which helps disseminate information throughout the network.

Also, Icthh (household ICT availability) shows relatively high centrality values for betweenness (0.520), closeness (0.605), and strength (0.948). Thus, it can be concluded that household ICT availability plays a significant role in ensuring effective digital engagement at both household and firm levels. In this case, digital infrastructure becomes a crucial factor in supporting the development of digital economic activity [132,133].

Innovsys and Gdpcpc have been identified as peripheral nodes with negative values for the analyzed centrality indicators. It suggests that innovation capability and income do not act as the direct factors responsible for the diffusion of online commerce but rather have an indirect influence through digital infrastructure and digital skills.

At that, telework has shown contradictory values in terms of centrality indicators, specifically, very high closeness (0.914) and negative betweenness (−0.728). In this regard, this variable reveals strong local influence but does not serve as a link within the network that enables the diffusion of economic activities based on digital skills.

As regards the dependent variable (Ecombiz), it shows relatively low centrality, suggesting a peripheral role and being mainly influenced by other variables.

Thus, in conclusion, the identified network reveals a significant role for digital inclusion in the diffusion of e-commerce. The central role of internet use and ICT availability

in mediating social-economic processes in relation to online business can be highlighted (Figure 10).



Figure 10. Network centrality measures of variables influencing e-commerce adoption.

The clustering coefficient analysis provides information about how each variable locally interacts within the network, indicating the degree to which nodes pack tightly together [134].The variable Ecombiz shows stable, high clustering values across all measures, particularly strong in the Onnela (0.924) and Zhang (1.316) indices, indicating that online sales business engagement is nested within a dense cluster of related digital and socioeconomic variables. This presents a strong interdependence between online shopping and the surrounding conditions of digital spread, in line with evidence that e-commerce systems are structurally bound by industrial and technological agglomeration [135]. The variable Intuse, for regular internet users, shows high clustering across the majority of measures, particularly under the Onnela index (1.306). This shows that internet use not only serves as a central variable but also co-exists within cohesive sub-structures of related variables, reinforcing bidirectional dependencies between connectivity and economic digitalization. Such behavior is consistent with hybrid data systems in online shopping, in which digital engagement supports inter-variable cohesion [136]. Conversely, Gdpcpc and Innovsys show negative clustering values across various metrics, indicating that income and innovation are located in the more isolated sections of the network, where their links with the other variables are weaker or less bidirectionally reinforcing. Theory in network topology underlines this explanation, as some socioeconomic indicators tend to reside in the peripheral sections of the closely knit digital settings. Icthh and Teleworking (Telework) show mixed results: Icthh shows poor clustering, while Teleworking (Telework) shows positive clustering in the Zhang measure. These findings indicate that home-based occupations

and digital household infrastructure contribute to localized association but play a more peripheral role in maintaining the overall network's cohesion. Similar structural tendencies were observed in theoretical models of clustering in graph products, where peripheral units can still maintain sub-network coherence [137]. In general, the findings confirm that online shopping activity and internet use form the most cohesive and bidirectionally reinforcing cluster in the digital economy, providing the structural anchors for the regional digitalization process in Italy (Figure 11).



Figure 11. Clustering coefficients of variables in the e-commerce adoption network.

In this regard, the weight matrix helps describe the strength and nature of the connections between variables, showing how all of them underpin the dynamic changes in e-commerce uptake [138,139]. The variable Ecombiz has very strong positive correlations with the indicators of Teleworking (Telework) (0.378) and ICTHH (0.313), showing the importance of digital household infrastructure and equipment as the key elements in the formation of digital business practices. In particular, the correlation indicates that the growth in the digitalization of household equipment and teleworking increases interest in digital business activities [140]. The negative correlation between Ecombiz and Innovsys (−0.202) indicates that innovations in the production system do not always translate into innovations in digital sales, due to various technological and industrial constraints [141]. As for the Teleworking (Telework) indicator, it shows very strong correlations with Innovsys (0.937) and Icthh (0.350), suggesting that innovations in teleworking are associated with technological progress. It means that a flexible working mode should be considered a part of the overall technological process [142]. On the other hand, the indicator of recurrent internet use (Intuse) shows low, insignificant correlations with other indicators. Specifically, the negative correlation with Icthh (−0.118) shows that individual computer use and family computer usage partially overlap. Finally, the majority of connections identified

by Gdpcpc indicate that per capita income has little bearing on the structure of digital economic interactions [134]. See Figure 12.

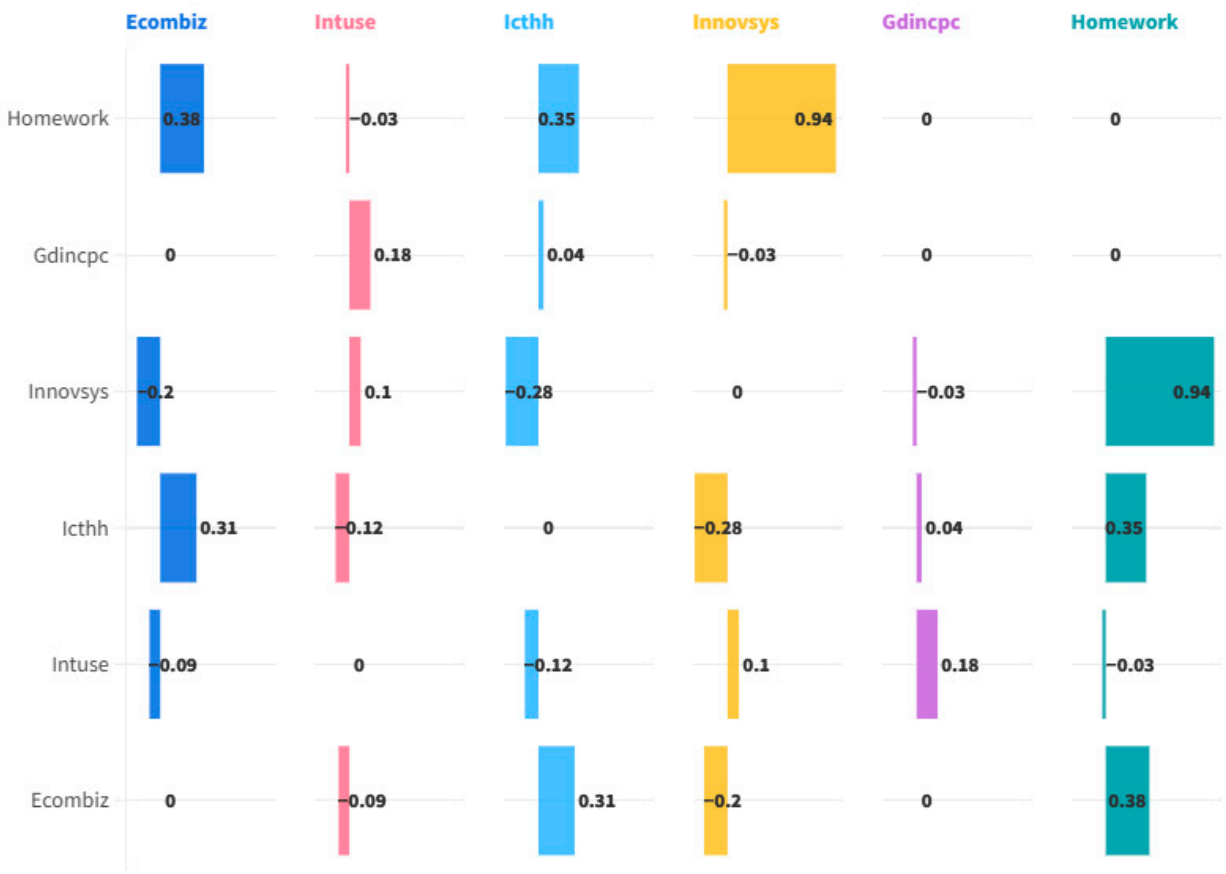


Figure 12. Weights matrix representing intervariable connections in the e-commerce adoption network.

This network graph illustrates the connections among variables that drive organizations' involvement in electronic commerce. As for the most prominent link, it can be observed between Intuse and Icthh, as these two variables have the highest association (0.94). This strong correlation demonstrates the significant interconnection between individual and household internet access, underscoring that connectivity underpins digital participation [143]. Taken together, the listed variables constitute the core technological factors that determine the spread of e-commerce [125].

As the dependent variable, Ecombiz shows a positive association with both Intuse (0.38) and Telework (0.31), indicating that higher internet usage and telework are associated with greater involvement in electronic commerce [144]. In addition, the positive association between Telework and Icthh (0.35) indicates that the presence of digital infrastructure enables remote work, thereby facilitating participation in the digital market [123].

At the same time, both Gdpcpc and Innovsys show weak correlations with the other variables, indicating their peripheral position in the described network. As such, low income and innovation potential do not directly affect e-commerce adoption but do so indirectly via digital infrastructure [145].

In general, the considered network indicates a well-connected digital ecosystem in which internet use, household connectivity, and participation in digital activities mutually influence one another. This result proves the key role of digital participation as the primary mechanism for the spread of e-commerce (Figure 13).

Centrality analysis provides a holistic overview of each variable's contribution to the network architecture and the relationships relevant to companies' involvement in online

commerce. Among all the variables used, Intuse (regular internet usage) receives the highest scores for strength and expected influence, thereby justifying its key place in the network. Specifically, this variable plays an integral role in the digital transformation process due to its direct and significant impact on other variables, particularly those related to connectivity and e-commerce activities [146,147]. Likewise, the ICT availability at households (Icthh) achieves relatively high centrality values in terms of strength and closeness, thus highlighting its critical intermediary function. In particular, it ensures the transition of digital effects into economic ones and, therefore, strengthens the link between connectivity and business conduct [148]. On the contrary, Ecombiz and Innovsys exhibit relatively low centrality scores across the board. This means they are located peripherally in the network, which can be explained by the fact that, even though Ecombiz serves as the dependent variable, it mostly reflects the existing situation rather than mediating it [149]. Additionally, Innovsys and Gdpcpc demonstrate low or even negative centrality values, indicating that their influence on e-commerce adoption is indirect and occurs through digital access and usage. Lastly, Telework can be described as a variable with a complex set of features, including high centrality in terms of closeness and moderate centrality scores overall. Therefore, it can be assumed that remote work affects the remaining variables; however, it is not a key channel for transmitting effects within the network [150]. Thus, the centrality structure highlights the critical importance of digital access and internet use in fostering network cohesion and the diffusion of e-commerce practices. Economic and innovation-related factors occupy a secondary position in the digital ecosystem. See Figure 14.

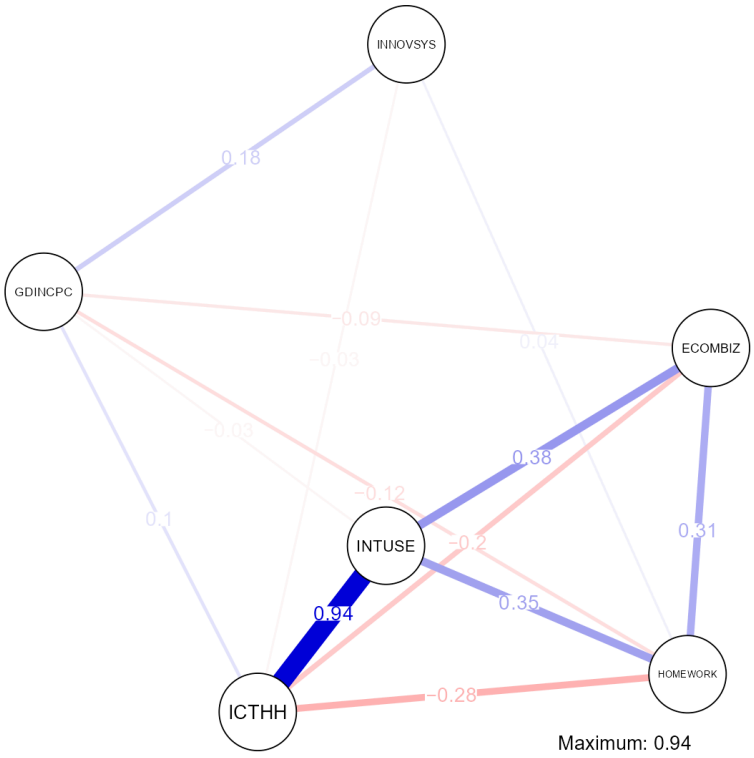


Figure 13. Network visualization of intervariable connections in e-commerce adoption.

The clustering graph represents the level of local aggregation of each node relative to others, based on the strength of association between them. Ecombiz shows consistently positive clustering in most cases, indicating that online business activities form a highly interrelated cluster of interdependent digital and socioeconomic phenomena [134]. This implies that companies engaging in electronic commerce operate in conditions where the use of digital resources, their development, and their integration into economic practices are

mutually reinforcing. Intuse shows positive clustering across all indices, particularly for the Onnela and WS indices, confirming that intensive internet usage serves as a driver that sustains the whole network through strong dependencies on other digital phenomena such as Icthh and Teleworking [151]. The current result proves that the structure of clustering plays a crucial role in understanding the phenomenon of propagation in the field of digital economy [152]. Icthh shows some degree of clustering, indicating that household connectivity enables local digital clusters to aggregate but does not act as an aggregator per se. In fact, Icthh serves as an infrastructural node for local aggregation, consistent with findings from generalized circulant networks, which show that infrastructure nodes tend to exhibit moderate levels of aggregation stability within a network system [153]. By contrast, Gdpcpc and Innovsys exhibit negative clustering across various indices, indicating that money capital and innovations are peripheral entities in the digital network, with limited impact through relatively weak interactions with other nodes [154]. Thus, these factors appear to operate more contextually than functionally in the network, having diffuse influence and little interaction. Finally, the Teleworking (Telework) index shows a mixed pattern of clustering, being positively clustered for some indices and less clustered for others; that is, remote work seems to be more interdependent in a local dimension with digital technologies than innovations and money capital. Overall, the current graph supports the hypothesis that internet usage and online business activities constitute the most local and aggregated elements in the digital network, as advanced clustering models for complex networks [134] suggest. See Figure 15.

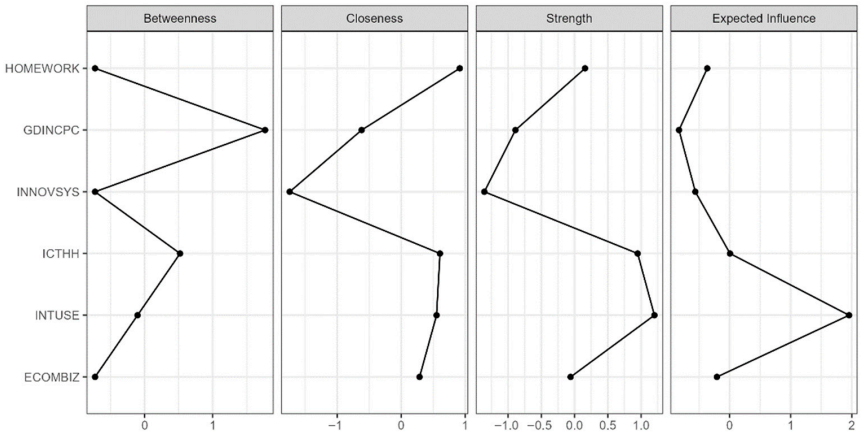


Figure 14. Centrality plot of network roles in e-commerce adoption.

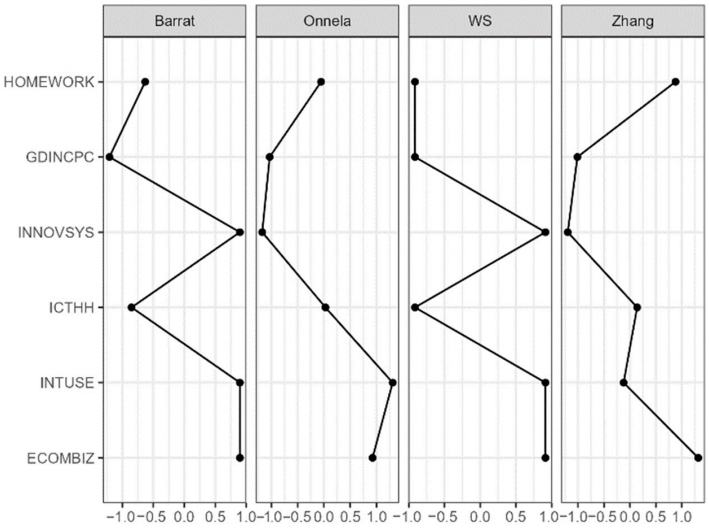


Figure 15. Clustering coefficient plot across four network measures (Barrat, Onnela, WS, Zhang).

9. Policy Implications for Enhancing Regional E-Commerce Adoption in Italy

It might be argued that adopting a bundle of regional economic policies in Italy could help foster the diffusion of web-based sales among SMEs selling to end customers. The policy recommendations proposed below are specifically oriented towards Italy's peculiarities, characterized by significant regional gaps in infrastructure, human resources, and capabilities, which affect the development of the digital economy. There are numerous empirical studies demonstrating the existence of the digital divide in Italy and the need to adopt appropriate policies to foster digital transformation in the country [155–157].

From this perspective, it is essential to acknowledge the importance of national strategies and policy-making processes for digital governance and public sector innovation, which can significantly contribute to the creation of an inclusive digital ecosystem [158,159]. Moreover, recent research shows that there are opportunities to adopt local policies to facilitate the further adoption of information technologies [95,160]. Therefore, it is critical to develop policies that are both economically rational and territorially oriented.

E-commerce Infrastructure Development. One of the core priorities in the field under discussion is the development of e-commerce infrastructure in Italy. Extending high-speed broadband network coverage in southern regions and rural municipalities is especially important, as it could promote a more uniform and inclusive use of the internet and electronic marketplaces [156,161]. At the same time, recent developments in the country show that improving digital infrastructure requires strengthening cybersecurity practices, payment systems, and the usability of platforms used by firms to promote their products and services [158,162].

Financial Incentives and Tax Breaks. Financial incentive measures and tax breaks are among the critical tools that can accelerate e-commerce adoption. It should be emphasized that such policies could also address challenges related to inequality and help SMEs engage in digitalization [163,164]. Thus, reducing entry barriers and providing opportunities for technological investment would help boost the competitiveness and innovation potential of businesses operating in regions with lower levels of development.

Access to Funding and Support. Finally, it is critical to create additional conditions for SMEs to access financing. Previous experience in the country indicates that public funding programs and territorial investment programs could help stimulate entrepreneurial activity [95,160]. In this regard, it is vital to provide additional support through advisory services and technical assistance to facilitate the implementation of solutions for digital commerce and marketing.

Training and Skill Development. There is a need to offer training programs and workshops to enhance relevant digital skills in e-commerce, helping businesses and the workforce prosper in the digital marketplace. Such training should allocate a reasonable share of topics to website development, digital marketing, and online sales management, providing participants with enough insight into the tools and strategies involved in digital e-commerce. Such programs must be industry- and business-specific, with varying sizes, so that whatever is taught remains relevant and addresses sector-specific challenges while seizing opportunities. With a bias towards practical learning, participants can put the skills and knowledge they have gained into practice, thereby making swift implementation a reality for e-commerce solutions in their operations. While this strategy improves the e-commerce readiness of businesses, it also enhances digital literacy among the workforce. By backing these training or skill development initiatives, governments and industry leaders would support a culture of continuous learning and adaptation in the fast-evolving digital economy [53,165]. In the Italian context, recent evidence highlights the importance of developing digital skills to reduce regional disparities and improve SME competitiveness [155].

Regional Collaboration and Networking. Such collaboration and networking among businesses, industry associations, and local government agencies would facilitate knowledge sharing and the adoption of best practices in e-commerce as a strategic approach to leveraging diverse skills toward propelling the digital economy forward. A regional e-commerce cluster or hub would establish a concentrated ecosystem where stakeholders can cooperate and exchange information easily. Such clusters act as incubators of innovation, where any business—regardless of size—can share resources, insights, and technology. The collaborative environment expedites the adoption of innovative practices in e-commerce and fosters a culture of continuous improvement and adaptability to emerging digital trends. These hubs can act as catalysts for regional economic development by attracting investment and talent and showcasing collective capabilities and success stories across the e-commerce sector. They provide a platform for government agencies to better engage with the business community in tailoring support services and regulatory frameworks that create an enabling environment for the growth of electronic commerce. Regional collaboration and networking can significantly reduce these start-up barriers in online markets and level the playing field for smaller enterprises, allowing them to compete at a larger scale [48,49,166].

Promotion of Digital Literacy. Spending on programs that improve digital literacy, not just for businesses but also for consumers, would go a long way toward creating an effective ecosystem for e-commerce. Businesses need to be enriched by the benefits they would reap from e-commerce: expanded market access, cost reductions, and enhanced insights into customer needs. By providing resources that ease business adoption—removing some of the traditional barriers, such as those that are inherently technological, security-related, and logistical—the policymakers and industry leaders can foster a more robust demand for electronic products and services. This approach will not only lead to faster digitalization of traditional businesses but also foster a more digitally well-informed clientele. Informed consumers who recognize the benefits and protections of e-commerce are more likely to participate, thereby driving demand upward through a positive cycle. Enhanced levels of digital literacy will eventually increase confidence in online platform use and further stimulate businesses to innovate and improve in the online sphere. In addition, these digital literacy efforts have helped to level the playing field for small and medium-sized enterprises with some of the more prominent players [4,58,167,168]. This is particularly relevant in Italy, where digital literacy gaps continue to affect both firms and consumers, especially in southern regions [156].

Streamlined Regulatory Environment. There is an urgent need to streamline and clarify the regulatory framework for e-commerce, including online transactions, personal data protection, and consumer protection, to facilitate the growth of digital commerce. Spelling out the friendly, enabling regulatory environment eliminates some relevant entry barriers or the great multiplication of business for those currently in operation and wishing to start or expand operations online. Eliminating unnecessary bureaucratic hurdles and creating a more predictable legal framework in the business sector will also foster greater confidence in planning and investment in e-commerce projects. This is especially important for small and medium-sized enterprises, which are often more burdened in complex regulatory settings. Clear rules—in both data protection and consumer rights—protect people and provide more confidence for consumers using online platforms. It becomes more likely that they will generate demand by engaging in e-commerce if consumers are assured of robust protection for their data and rights in digital transactions. So, simplifying regulations actually benefits businesses and creates a cleaner e-commerce environment where trust, innovation, and access generate mutual benefit between enterprises and consumers. Such an environment would be conducive to investment, innovation, and growth in the digital

economy, which is essential for building a robust e-commerce sector [39,169]. In Italy, recent studies emphasize the role of regulatory simplification and digital governance reforms in supporting the transition toward a more efficient and inclusive digital economy [159].

Innovative Contribution. The analysis of the presence of firms using e-commerce technologies across Italian regions is very important, given the significant gap between Central-Northern and Southern Italy. The economic gap created by unification in 1861, although the object of economic policies aimed at reducing it, further widened the differences between Italian regions. These differences also became cultural and influenced the distribution of human capital at the regional level. Consequently, there are also regional disparities in the use of technologies, underscoring the problem of some areas being unable to effectively access and use new technologies, such as online sales. Another important point is that the Italian case is by no means an anomaly and that every nation necessarily contains heterogeneity between its regions. Indeed, most countries have regions that are falling behind others. The Italian case can therefore serve as a model even for other countries with large regional differences in firms' capacity to adopt new technologies.

Another innovative contribution is that the analysis considers variables closely related to the dimensions of technological innovation, human capital, and the knowledge economy. In this line of thinking, a model is proposed that goes beyond the classic analyses based on per capita GDP or the degree of investment in research and development and technological innovation. This way, one can reach a model that can be used to separate the impact of innovation orientation on the use of online sales tools by SMEs. Therefore, the determinants of technological innovation and human capital are particularly congruent with this analysis since the role of information technologies is linked to the population's capacity to develop a dynamic and creative approach to the knowledge economy. This topic becomes particularly relevant, above all in the current international scenario, characterized by competition between the United States and China in the technological field. This would mean that, within such a framework, relations of a metric type should be sought that can stimulate economic policies to guide SMEs toward the widespread adoption of new technologies. The article, therefore, makes a relevant, innovative contribution by choosing the issue of regional disparity for analysis, which is becoming increasingly important in the current rediscovery of the geopolitical role of economic areas in the international context. A further innovative contribution consists of the analysis of determinants of human and knowledge capital that will enable SMEs to be more active in experimenting with new technologies in their business activities.

10. Conclusions

This research provides valuable regional insights into the diffusion dynamics of e-commerce among Italian SMEs by combining econometrics, machine learning, and network science within a single methodology. The main contribution of this paper is to combine analyses of causal inference, prediction, and systemic networks to investigate digitalization as a multifaceted process shaped by interactions among technological, socio-economic, and institutional dimensions. Empirically, the analysis suggests that the main drivers of e-commerce diffusion across Italian regions are the Intuse (internet use) and Icthh (household ICT availability) variables, whereas income and innovation levels exert no direct impact. It means that the higher level of digital participation and engagement in e-commerce is positively associated. However, infrastructure itself cannot be viewed as a sufficient condition of diffusion because behavioral, social, and institutional preconditions should also be considered. Concerning methodology, the econometric models ensure causal inference and address endogeneity between digitalization and economic development variables. Machine learning algorithms capture nonlinear relationships and provide stable variable importance

across different models. Finally, the k-means algorithm, optimized for clustering quality using Silhouette coefficients, reveals two types of Italian regions: more developed northern and central regions, and lagging but developing southern and insular regions. As can be seen, the North–South digital gap persists, but convergence in this field is also underway due to remote work, digital business operations, and government investments. Network science adds a systemic dimension to the analysis because both digital participation and ICT access are identified as the most central nodes in the network, serving as the mechanisms through which connections between economic and social variables occur. Innovation and income are considered less important for e-commerce diffusion, meaning that this process depends more on digital participation and technological development. In terms of general significance, this research is relevant to countries with existing regional differences. They may include both advanced and developing economies such as Brazil, India, South Africa, Turkey, the UK, Spain, Germany, the USA, and Mexico. In such countries, digitalization serves as a catalyst for regional development by facilitating market access, fostering local entrepreneurship, and increasing the competitiveness of lagging areas. From the perspective of Italy's National Recovery and Resilience Plan (PNRR), this research provides insights into how regional competitiveness can be ensured through effective policies focused on the digital environment. Methodologically, this research contributes to the literature by combining instrumental-variable panel-data estimation, clustering algorithms, machine learning, and network analysis. Overall, it may be concluded that in terms of regional competitiveness, Italy and other fragmented economies do not depend on income and infrastructure as much as they rely on digital participation and human capital development.

Author Contributions: Conceptualization, A.L., C.D., A.C. and M.A.; methodology, A.L., C.D., A.C. and M.A.; software, A.L., C.D., A.C. and M.A.; validation, A.L., C.D., A.C. and M.A.; formal analysis, A.L., C.D., A.C. and M.A.; investigation, A.L., C.D., A.C. and M.A.; resources, A.L., C.D., A.C. and M.A.; data curation, A.L., C.D., A.C. and M.A.; writing—original draft preparation, A.L., C.D., A.C. and M.A.; writing—review and editing, A.L., C.D., A.C. and M.A.; visualization, A.L., C.D., A.C. and M.A.; supervision, A.L., C.D., A.C. and M.A.; project administration, A.L., C.D., A.C. and M.A.; funding acquisition, A.L., C.D., A.C. and M.A. All authors have read and agreed to the published version of the manuscript.

Funding: The authors declare that no funds, grants, or other financial support were received during the preparation of this manuscript.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are openly available in [ISTAT-BES] [https://www.istat.it/statistiche-per-temi/focus/benessere-e-sostenibilita/la-misurazione-del-benessere-bes/gli-indicatori-del-bes/] (accessed on 21 April 2026).

Conflicts of Interest: The authors have no relevant financial or non-financial interests to disclose. We affirm that there are no personal, professional, or financial conflicts of interest related to this work.

Appendix A. Descriptive Statistics

Table A1. Descriptive statistics summary for key variables influencing e-commerce adoption.

	Ecombiz	Intuse	Icthh	Innovsys	Gdpcpc	Teleworking (Telework)
Valid	380	380	380	380	380	380
Missing	0	0	0	0	0	0
Mode	0.000	0.000	0.000	0.000	0.000	0.000

Table A1. *Cont.*

	Ecombiz	Intuse	Icthh	Innovsys	Gdpcpc	Teleworking (Telework)
Median	3.550	50.350	55.200	0.000	18.350	0.000
Mean	5.124	48.325	51.746	19.453	16.735	2.366
Std. Error of Mean	0.317	0.986	0.874	1.194	266.737	0.235
95% CI Mean Upper	5.747	50.263	53.465	21.801	17.260	2.828
95% CI Mean Lower	4.501	46.387	50.028	17.105	16.211	1.903
Std. Deviation	6.175	19.213	17.035	23.279	5.199	4.588
95% CI Std. Dev. Upper	6.649	20.686	18.341	25.063	5.598	4.939
95% CI Std. Dev. Lower	5.765	17.938	15.904	21.733	4.854	4.283
Coefficient of variation	1.205	0.398	0.329	1.197	0.311	1.939
MAD	3.550	14.000	10.100	0.000	3.178	0.000
MAD robust	5.263	20.756	14.974	0.000	4.712	0.000
IQR	8.900	27.900	21.225	44.325	6.796	3.150
Variance	38.134	369.159	290.196	541.913	$2.704 \times 10^{+7}$	21.048
95% CI Variance Upper	44.204	427.919	336.388	628.171	$3.134 \times 10^{+7}$	24.398
95% CI Variance Lower	33.238	321.760	252.937	472.334	$2.357 \times 10^{+7}$	18.345
Skewness	1.233	−0.661	−1.395	0.432	−1.587	1.964
Std. Error of Skewness	0.125	0.125	0.125	0.125	0.125	0.125
Kurtosis	1.420	0.033	2.085	−1.680	3.086	3.087
Std. Error of Kurtosis	0.250	0.250	0.250	0.250	0.250	0.250
Shapiro–Wilk	0.811	0.951	0.872	0.714	0.831	0.589
p-value of Shapiro–Wilk	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Range	30.900	81.500	77.500	62.400	24.324	23.000
Minimum	0.000	0.000	0.000	0.000	0.000	0.000
Maximum	30.900	81.500	77.500	62.400	24.324	23.000
25th percentile	0.000	35.700	42.825	0.000	13.660	0.000
50th percentile	3.550	50.350	55.200	0.000	18.350	0.000
75th percentile	8.900	63.600	64.050	44.325	20.457	3.150
Sum	1.947	18.363	19.663	7.392	$6.360 \times 10^{+6}$	898.900

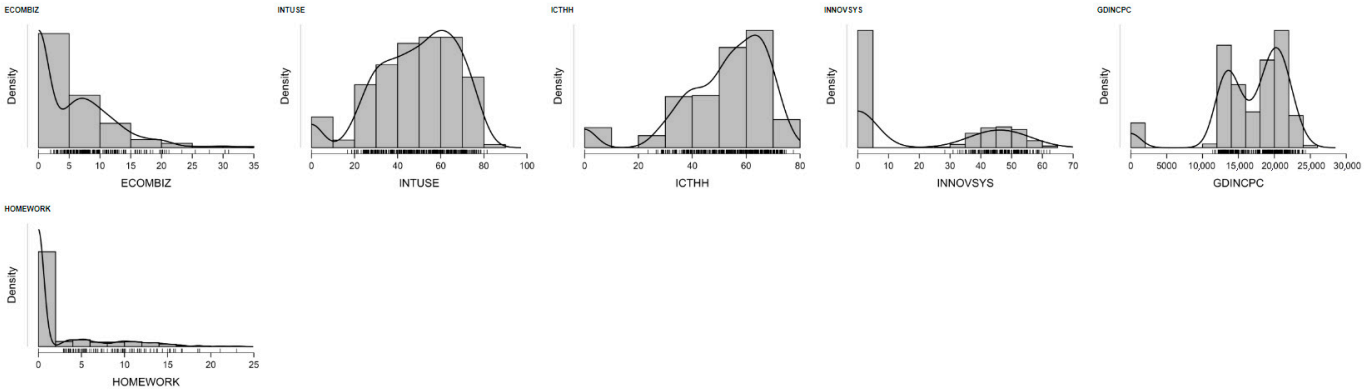


Figure A1. Distribution plots of variables related to e-commerce adoption.

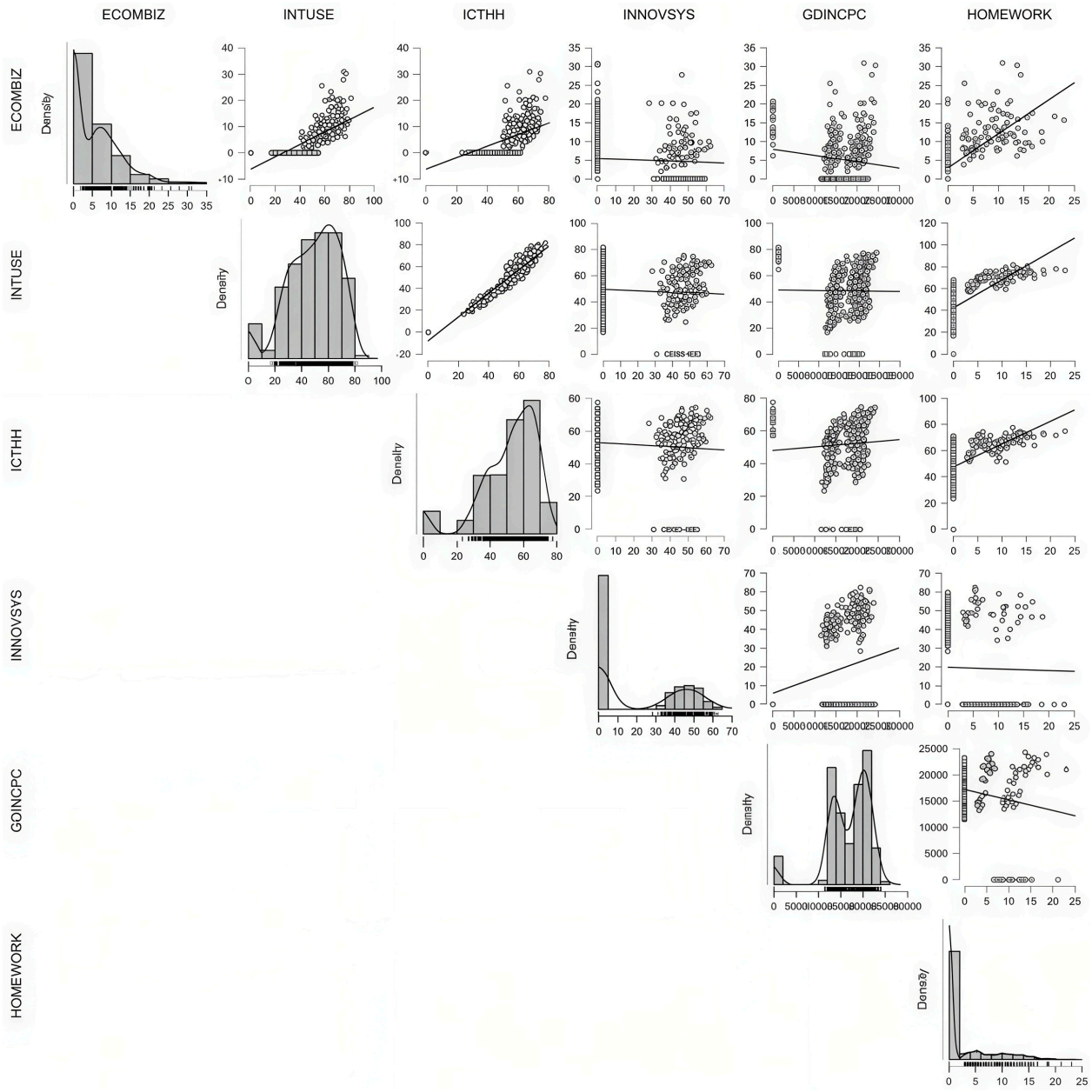


Figure A2. Pairwise scatterplot matrix and density distributions of e-commerce adoption variables.

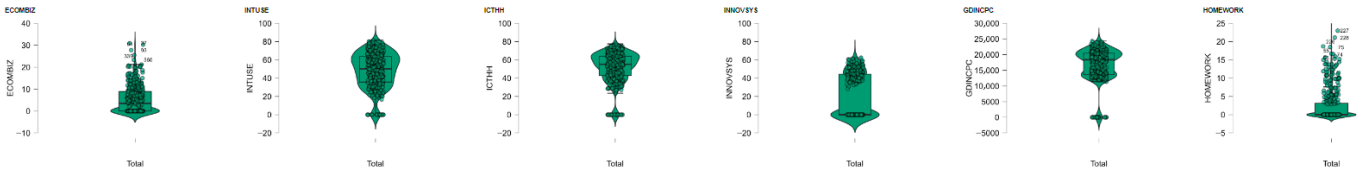


Figure A3. Violin plots of variable distributions for e-commerce adoption analysis.

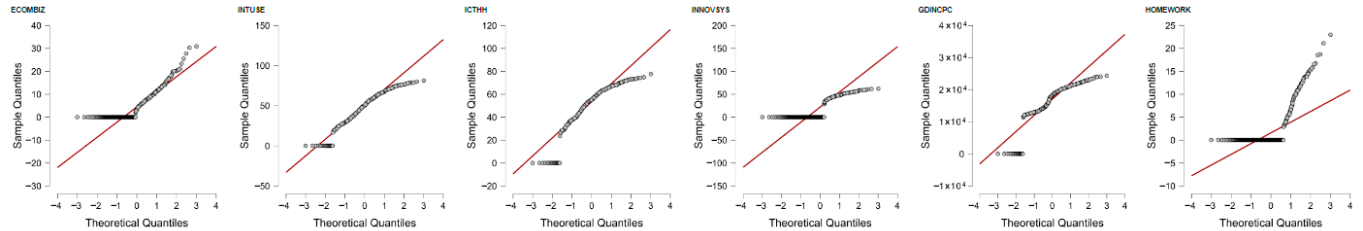


Figure A4. Q-Q plots assessing normality of key variables.

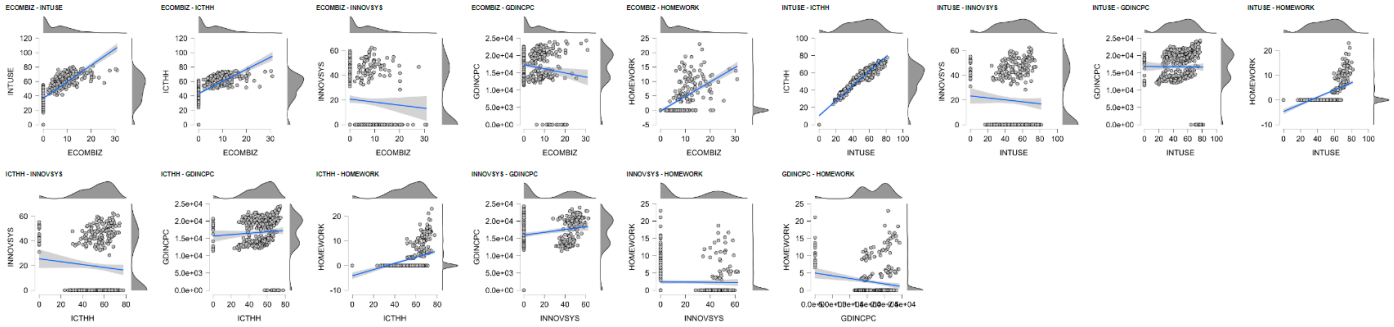


Figure A5. Pairwise scatterplots with density plots and regression lines of e-commerce adoption variables.

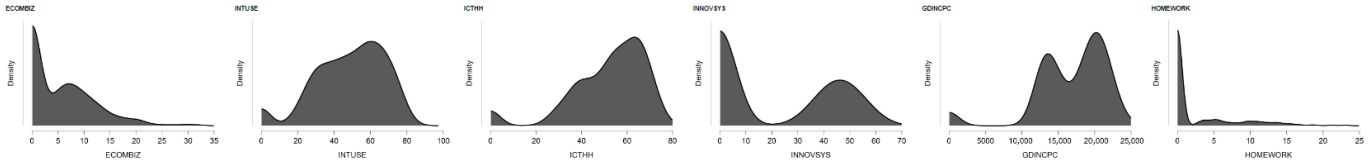


Figure A6. Kernel density estimates of e-commerce adoption and digital variables.

Table A2. Covariance matrix of e-commerce and digital variables.

Covariance	Ecombiz	Intuse	Icthh	Innovsys	Gdpcpc	Teleworking (Telework)
Ecombiz	38.134	86.991	64.102	−9.325	−4.540	19.126
Intuse	86.991	369.159	315.380	−29.048	−947.207	53.735
Icthh	64.102	315.380	290.196	−34.032	5.856	36.504
Innovsys	−9.325	−29.048	−34.032	541.913	21.802	−1.607
Gdpcpc	−4.540	−947.207	5.856	21.802	$2.704 \times 10^{+7}$	−4.230
Teleworking (Telework)	19.126	53.735	36.504	−1.607	−4.230	21.048

Table A3. Correlation matrix of e-commerce and digital variables.

Correlation	Ecombiz	Intuse	Icthh	Innovsys	Gdpcpc	Teleworking (Telework)
Ecombiz	1.000	0.733	0.609	−0.065	−0.141	0.675
Intuse	0.733	1.000	0.964	−0.065	−0.009	0.610
Icthh	0.609	0.964	1.000	−0.086	0.066	0.467
Innovsys	−0.065	−0.065	−0.086	1.000	0.180	−0.015
Gdpcpc	−0.141	−0.009	0.066	0.180	1.000	−0.177
Teleworking (Telework)	0.675	0.610	0.467	−0.015	−0.177	1.000

References

1. Hasan, A.; Amrusi; Musfiana; Mardhani, M. An overview of e-commerce adoption in Indonesian SMEs. *J. Phys. Conf. Ser.* **2021**, *1811*, 012104. [CrossRef]
2. Tong, A.; Gong, R. *The Impact of COVID-19 on SME Digitalisation in Malaysia*; The London School of Economics and Political Science: London, UK, 2020.
3. Alimonti, R.; Mautino, L.; Stammati, L. E-commerce growth: Competition and regulatory implications for the postal sector. In *The Changing Postal Environment: Market and Policy Innovation*; Springer International Publishing: Cham, Switzerland, 2020; pp. 167–181.
4. D'Adamo, I.; González-Sánchez, R.; Medina-Salgado, M.S.; Settembre-Blundo, D. E-commerce calls for cyber-security and sus-tainability: How european citizens look for a trusted online environment. *Sustainability* **2021**, *13*, 6752. [CrossRef]
5. Calignano, G.; Quarta, C.A. The persistence of regional disparities in Italy through the lens of the European Union nanotech-nology network. *Reg. Stud. Reg. Sci.* **2015**, *2*, 470–479.
6. Marino, A.; Pariso, P.; Picariello, M. Exploring the Economic Recovery of Italy's Regions Post-COVID-19: A focus on Energy, Services, ICT Opportunities, and the Digital Divide. *Int. J. Energy Econ. Policy* **2023**, *13*, 271–280. [CrossRef]

7. Duan, H.; Dawod, A.Y.; Wan, G. The personalization paradox in AI-driven tourism e-commerce: Psychological reactance, threat-substitution, and the moderating role of privacy concerns. *J. Theor. Appl. Electron. Commer. Res.* **2026**, *21*, 127. [CrossRef]
8. Bunea, O.I.; Corboș, R.A. When does artificial intelligence pay off in electronic retailing? A dual-path model from implemen-tation to competitive advantage. *J. Theor. Appl. Electron. Commer. Res.* **2026**, *21*, 119. [CrossRef]
9. Lee, J.-M.; Kim, S. When choice disappears: Perceived coercion and avoidance of self-service technologies among older consumers. *J. Theor. Appl. Electron. Commer. Res.* **2026**, *21*, 126. [CrossRef]
10. Wang, C.-C.; Hsu, Y.-T.; Kuo, H.-Y. Enhancing supply chain resilience in textile SMEs: A human-centric custom-er-to-manufacturer framework using public e-commerce data. *J. Theor. Appl. Electron. Commer. Res.* **2026**, *21*, 123. [CrossRef]
11. Hussain, A.; Akbar, M.; Shahzad, A.; Poulova, P.; Akbar, A.; Hassan, R. E-commerce and SME performance: The moderating influence of entrepreneurial competencies. *Adm. Sci.* **2022**, *12*, 13. [CrossRef]
12. Kim, K.; Lim, G. International dynamic marketing capabilities of emerging-market small business on e-commerce. *J. Theor. Appl. Electron. Commer. Res.* **2022**, *17*, 199–211. [CrossRef]
13. Hussain, A.; Shahzad, A.; Hassan, R. Organizational and environmental factors with the mediating role of e-commerce and SME performance. *J. Open Innov. Technol. Mark. Complex.* **2020**, *6*, 196. [CrossRef]
14. Yang, T.; Xun, J.; Chong, W.K. Complementary resources and SME firm performance: The role of external readiness and E-commerce functionality. *Ind. Manag. Data Syst.* **2022**, *122*, 1128–1151. [CrossRef]
15. Pan, L.; Fu, X.; Li, Y. SME participation in cross-border e-commerce as an entry mode to foreign markets: A driver of innova-tion or not? *Electron. Commer. Res.* **2023**, *23*, 2327–2356. [CrossRef]
16. Syafrizal, M. Web-Based SME Online Marketing System (E-Commerce). *Int. J. Softw. Eng. Comput. Sci. (IJSECS)* **2021**, *1*, 75–79. [CrossRef]
17. Almtiri, Z.; Miah, S.J.; Noman, N. Application of E-commerce Technologies in Accelerating the Success of SME Operation. In *Proceedings of Seventh International Congress on Information and Communication Technology*; Springer: Singapore, 2023; pp. 463–470.
18. Guo, J.; Jia, F.; Yan, F.; Chen, L. E-commerce supply chain finance for SMEs: The role of green innovation. *Int. J. Logist. Res. Appl.* **2023**, *27*, 1596–1615. [CrossRef]
19. Cheong, J.Q.; Lee, N.F.C.; Fadzlee, M.; Mansur, K.H.M. A systematic literature review of COVID-19 impact to sme's adoption of e-commerce. *J. BIMP-EAGA Reg. Dev.* **2020**, *6*, 19–33. [CrossRef]
20. Ismail, N.A.; Masud, M.M. Prospects and challenges in improving e-commerce connectivity in Malaysia. In *E-Commerce Connectivity in ASEAN*; Economic Research Institute for ASEAN and East Asia: Jakarta, Indonesia, 2020; p. 78.
21. Özbek, N.; Hånell, S.M.; Tolstoy, D.; Nordman, E.R. Exploring different responses to mimetic pressures: An institutional theory perspective on e-commerce adoption of an internationalizing retail SME. *Int. Rev. Retail Distrib. Consum. Res.* **2024**, *34*, 14–32. [CrossRef]
22. Rawash, H.N. E-commerce adopting TOE model by SMEs in Jordan. *Multicult. Educ.* **2021**, *7*, 118–122.
23. Algieri, B.; Álvarez, A. Assessing the ability of regions to attract foreign tourists: The case of Italy. *Tour. Econ.* **2023**, *29*, 788–811. [CrossRef]
24. Franch, M.; Irimiás, A.; Buffa, F. Place identity and war heritage: Managerial challenges in tourism development in Trentino and Alto Adige/Südtirol. *Place Brand. Public Dipl.* **2017**, *13*, 119–135. [CrossRef]
25. Bentivogli, C.; Ferraresi, T.; Monti, P.; Paniccià, R.; Rosignoli, S. Italian regions in global value chains: An input-output approach. *Politica Econ.* **2019**, *35*, 55–94. [CrossRef]
26. Billé, A.G.; Tomelleri, A.; Ravazzolo, F. Forecasting regional GDPs: A comparison with spatial dynamic panel data models. *Spat. Econ. Anal.* **2023**, *18*, 530–551. [CrossRef]
27. Brunetta, G.; Morandi, C.; Tamini, L. Nuovi formati commerciali e polaritá interregionali: Lombardía, Piemonte, Emilia Romagna. *Ciudad. Rev. Inst. Univ. Urbanística Univ. Valladolid* **2007**, *10*, 203–212.
28. Burlina, C.; Casadei, P.; Crociata, A. Economic complexity and firm performance in the cultural and creative sector: Evidence from Italian provinces. *Eur. Urban Reg. Stud.* **2023**, *30*, 152–171. [CrossRef]
29. Carmela, G. Development Opportunities and Outskirts. The Innovative Service Sector in the South of Italy. *Riv. Econ. Mezzog.* **2020**, *4*, 781–812.
30. Orusa, T.; Mondino, E.B. Exploring short-term climate change effects on rangelands and broad-leaved forests by free satellite data in Aosta Valley (Northwest Italy). *Climate* **2021**, *9*, 47. [CrossRef]
31. Vergamini, D.; Bartolini, F.; Prosperi, P.; Brunori, G. Explaining regional dynamics of marketing strategies: The experience of the Tuscan wine producers. *J. Rural Stud.* **2019**, *72*, 136–152. [CrossRef]
32. Podestà, F. The economic impact of the Friuli-Venezia Giulia autonomy: A synthetic control analysis of asymmetric Italian federalism. *Ann. Reg. Sci.* **2017**, *58*, 21–37. [CrossRef]
33. Ardiansah, M.N.; Chariri, A.; Raharja, S. Does E-Commerce Impact on Accounting Information System and Organizations Performance? In *Proceedings of the 2nd Annual Conference on Blended Learning, Educational Technology and Innovation (ACBLETI 2020)*; Atlantis Press: Dordrecht, The Netherlands, 2021; pp. 476–481.

34. Barata, S.F.P.G.; Ferreira, F.A.F.; Carayannis, E.G.; Ferreira, J.J.M. Determinants of E-Commerce, Artificial Intelligence, and Agile Methods in Small-and Medium-Sized Enterprises. *IEEE Trans. Eng. Manag.* **2023**, *71*, 6903–6917. [CrossRef]
35. Barkatullah, A.H. Does self-regulation provide legal protection and security to e-commerce consumers? *Electron. Commer. Res. Appl.* **2018**, *30*, 94–101. [CrossRef]
36. Caputo, F.; Fiano, F.; Riso, T.; Romano, M.; Maalaoui, A. Digital platforms and international performance of Italian SMEs: An exploitation-based overview. *Int. Mark. Rev.* **2022**, *39*, 568–585. [CrossRef]
37. Chang, S.; Li, A.; Wang, X.; Zhang, J. Optimal combination of platform channel contract and guarantee financing strategy in e-commerce market. *Transp. Res. Part E Logist. Transp. Rev.* **2023**, *172*, 103094. [CrossRef]
38. Chen, C.-L.; Lin, Y.-C.; Chen, W.-H.; Chao, C.-F.; Pandia, H. Role of government to enhance digital transformation in small service business. *Sustainability* **2021**, *13*, 1028. [CrossRef]
39. Chawla, N.; Kumar, B. E-commerce and consumer protection in India: The emerging trend. *J. Bus. Ethics* **2022**, *180*, 581–604. [CrossRef]
40. Chotimah, C.; Sutikno; Setiawan. Modelling of income inequality in east java using geographically weighted panel regression. In *IOP Conference Series: Materials Science and Engineering*; IOP Publishing: Bristol, UK, 2019; Volume 546, p. 052019.
41. Costa, J.; Castro, R. SMEs must go online—E-commerce as an escape hatch for resilience and survivability. *J. Theor. Appl. Electron. Commer. Res.* **2021**, *16*, 3043–3062. [CrossRef]
42. Dai, H.; Xiao, Q.; Yan, N.; Xu, X.; Tong, T. What Influences Online Sales Across Different Types of E-Commerce Platforms. *Int. J. Electron. Commer.* **2022**, *26*, 311–330. [CrossRef]
43. Avdeeva, V.M. Tax incentives to encourage the implementation of digital innovations. *Nalogi Nalogoobloz. = Taxes Tax.* **2021**, *6*, 1–11. [CrossRef]
44. Awe, Y.N.; Ertemel, A.V. Enhancement of micro small businesses in the gambia through digitalization: Investigating youth entrepreneurs perception, use and inhibitor of e-commerce technology. *İSTANBUL Ticaret ÜNiversitesi Work. Pap. Ser.* **2021**, *2*, 25–42.
45. Cirillo, V.; Fanti, L.; Mina, A.; Ricci, A. New digital technologies and firm performance in the Italian economy. *Ind. Innov.* **2023**, *30*, 159–188. [CrossRef]
46. Dinis, A.; Lemos, K.; Serra, S. Tax incentives for SMEs' digital transformation. In *Proceedings of the 2023 18th Iberian Conference on Information Systems and Technologies (CISTI)*; IEEE: Piscataway, NJ, USA, 2023; pp. 1–6.
47. Elia, S.; Giuffrida, M.; Mariani, M.M.; Bresciani, S. Resources and digital export: An RBV perspective on the role of digital technologies and capabilities in cross-border e-commerce. *J. Bus. Res.* **2021**, *132*, 158–169. [CrossRef]
48. Frick, J. Establishing Effective Business/Stakeholder Networks to Underpin Regional Digital Strategies. *Int. J. Bus. Manag.* **2023**, *18*, 63–68. [CrossRef]
49. Georgescu, A.; Peter, M.K.; Avasilcai, S. A business ecosystem framework for SME development through associative and non-associative business structures in the digital age. *Cogent Bus. Manag.* **2022**, *9*, 2143310. [CrossRef]
50. Kadir, S.; Shaikh, J.M. The effects of e-commerce businesses to small-medium enterprises: Media techniques and technology. In *AIP Conference Proceeding*; AIP Publishing: Melville, NY, USA, 2023; Volume 2643.
51. Khodjayev, A.R. Efficiency of using modern information and communication technologies in small business. In *World Science: Problems and Innovations*; Perfect Publishing: Toronto, ON, Canada, 2021; pp. 130–132.
52. Patti, A.; Schifilliti, V. The effects of innovation and digitalization on business dynamism: An empirical study from Italian regions. *J. Econ. Bus. Lett.* **2023**, *3*, 1–8. [CrossRef]
53. Arianti, L. Digital Literacy Campaign to Improve the Community's Economy. *J. PKM Manaj. Bisnis* **2023**, *3*, 75–84. [CrossRef]
54. Gallardo, R. Bringing communities into the digital age. *State Local Gov. Rev.* **2019**, *51*, 233–241. [CrossRef]
55. Gutiérrez-Ángel, N.; Sánchez-García, J.-N.; Mercader-Rubio, I.; García-Martín, J.; Brito-Costa, S. Digital literacy in the university setting: A literature review of empirical studies between 2010 and 2021. *Front. Psychol.* **2022**, *13*, 896800. [CrossRef]
56. Di Pietro, G. Changes in Italy's education-related digital divide. *Econ. Aff.* **2021**, *41*, 252–270. [CrossRef]
57. Lestari, D.; Za, S.Z.; Maria, S.; Wardhani, W.; Yudaruddin, R. The impact of COVID-19 pandemic on performance of small enterprises that are e-commerce adopters and non-adopters. *Probl. Perspect. Manag.* **2021**, *19*, 467. [CrossRef]
58. Steyn, R.A. Changing thoughts towards digital literacy interventions for South African entrepreneurs. *Read. Writ. J. Read. As-soc. S. Afr.* **2018**, *9*, 1–9. [CrossRef]
59. van Kessel, R.; Wong, B.L.H.; Rubinić, I.; O'Nuallain, E.; Czabanowska, K. Is Europe prepared to go digital? making the case for developing digital capacity: An exploratory analysis of Eurostat survey data. *PLoS Digit. Health* **2022**, *1*, e0000013. [CrossRef]
60. Natale, M.T.; Piccininno, M. Digital cultural heritage and tourism recommendations for cultural institutions. *Uncommon Cult.* **2015**, *6*, 52–64.
61. Fox, C.S.; Hwang, S.; Nieto, K.; Valentino, M.; Mutalik, K.; Massaro, J.M.; Benjamin, E.J.; Murabito, J.M. Digital connectedness in the Framingham heart study. *J. Am. Heart Assoc.* **2016**, *5*, e003193. [CrossRef] [PubMed]

62. Ashari, I.F.; Nugroho, E.D.; Baraku, R.; Yanda, I.N.; Liwardana, R. Analysis of elbow, silhouette, Davies-Bouldin, Ca-linski-Harabasz, and rand-index evaluation on k-means algorithm for classifying flood-affected areas in Jakarta. *J. Appl. Inform. Comput.* **2023**, *7*, 95–103. [CrossRef]
63. Belkadi, A.; Kenouche, S.; Melkemi, N.; Daoud, I.; Djebaili, R. K-means clustering analysis, ADME/pharmacokinetic prediction, MEP, and molecular docking studies of potential cytotoxic agents. *Struct. Chem.* **2021**, *32*, 2235–2249. [CrossRef]
64. Bentéjac, C.; Csörgő, A.; Martínez-Muñoz, G. A comparative analysis of gradient boosting algorithms. *Artif. Intell. Rev.* **2021**, *54*, 1937–1967. [CrossRef]
65. Chaurasia, V.; Pal, S. Applications of machine learning techniques to predict diagnostic breast cancer. *SN Comput. Sci.* **2020**, *1*, 270. [CrossRef]
66. Dasig, D.D.; Calantoc, D.J.R.; Guarin, R.V.F.; Taduyo, M.A.B.; Ferrer, C.N.; Claricia, E.E.; Agus, G.E. Predicting Customer Purchase Decisions using Data Mining Technique. In *Proceedings of the 2022 IEEE 14th International Conference on Humanoid, Nanotechnology, In-formation Technology, Communication and Control, Environment, and Management (HNICEM)*; IEEE: Piscataway, NJ, USA, 2022.
67. de Araujo, M.H.; Reinhard, N. Substituting computers for mobile phones? An analysis of the effect of device divide on digital skills in Brazil. In *Proceedings of the Electronic Participation: 11th IFIP WG 8.5 International Conference, ePart 2019, San Benedetto Del Tronto, Italy, 2–4 September 2019*; Springer International Publishing: Cham, Switzerland, 2019; Volume 11, pp. 142–154.
68. Dinh, D.T.; Fujinami, T.; Huynh, V.N. Estimating the optimal number of clusters in categorical data clustering by silhouette coefficient. In *Proceedings of the Knowledge and Systems Sciences: 20th International Symposium KSS*; Springer: Singapore, 2019; Volume 20, pp. 1–17.
69. Leogrande, A.; Costantiello, A.; Laureti, L. k-Means Clusterization and Machine Learning Prediction of European Most Cited Scientific Publications. *Int. J. Entrep.* **2022**, *27*, 4195846. [CrossRef]
70. Mohapatra, S.; Mukherjee, R.; Roy, A.; Sengupta, A.; Puniyani, A. Can ensemble machine learning methods predict stock re-turns for Indian banks using technical indicators? *J. Risk Financ. Manag.* **2022**, *15*, 350. [CrossRef]
71. Nabipour, M.; Nayyeri, P.; Jabani, H.; Mosavi, A. Predicting stock market trends using machine learning and deep learning algorithms via continuous and binary data; a comparative analysis. *IEEE Access* **2020**, *8*, 150199–150212. [CrossRef]
72. Patel, P.; Sivaiah, B.; Patel, R. Approaches for finding optimal number of clusters using k-means and agglomerative hierarchical clustering techniques. In *Proceedings of the International Conference on Intelligent Controller and Computing for Smart Power (ICICCSP)*; IEEE: Piscataway, NJ, USA, 2022; pp. 1–6.
73. Punhani, A.; Faujdar, N.; Mishra, K.K.; Subramanian, M. Binning-based silhouette approach to find the optimal cluster using K-means. *IEEE Access* **2022**, *10*, 115025–115032. [CrossRef]
74. Zhou, W. K-Means Clustering Algorithm Analysis on Specific Economic Development Problems in Target Countries. In *Proceedings of the 2021 2nd International Conference on Computer Science and Management Technology (ICCSMT)*; IEEE: Piscataway, NJ, USA, 2021; pp. 396–402.
75. Bellandi, M. Some Notes on the Impacts of COVID-19 on Italian SME Productive Systems. *Symphonya* **2020**, *2*, 63–72. [CrossRef]
76. Harto, B.; Komalasari, R. Optimalisasi platform online internet marketing untuk SME Little Rose Bandung. *Empower. Community* **2020**, *1*, 1–6. [CrossRef]
77. Penco, L.; Profumo, G.; Serravalle, F.; Viassone, M. Has COVID-19 pushed digitalisation in SMEs? The role of entrepreneurial orientation. *J. Small Bus. Enterp. Dev.* **2022**, *30*, 311–341. [CrossRef]
78. Priambodo, I.T.; Sasmoko, S.; Abdinagoro, S.B.; Bandur, A. E-Commerce readiness of creative industry during the COVID-19 pandemic in Indonesia. *J. Asian Financ. Econ. Bus.* **2021**, *8*, 865–873.
79. Sun, M. The internet and SME participation in exports. *Inf. Econ. Policy* **2021**, *57*, 100940. [CrossRef]
80. Ilham, M.; Ratnamiasih, I. E-Commerce and Competence, Impact on Business Performance: Empirical Study on Implementa-tion E-Commerce by Students at Pasundan University. *J. Account. Financ. Tax. Audit. (JAFTA)* **2021**, *3*, 19–32. [CrossRef]
81. Ahmad, M.; Siraj, S. Determinants Impacting the Assimilation Stages of E-commerce in SMEs: A Modified TOE Framework. *Int. J. Electron. Commer. Stud.* **2023**, *14*, 101–126. [CrossRef]
82. Hamida, A.G. Exploring the Adoption of E-Commerce among Small and Medium Enterprises in Saudi Arabia: A Technology-Organization-Environment Perspective. *Digit. Manag. Sci. J.* **2024**, *2*, 1–17. [CrossRef]
83. Loo, M.K.; Ramachandran, S.; Yusof, R.N.R. Systematic review of factors and barriers influencing e-commerce adoption among SMEs over the last decade: A TOE framework perspective. *J. Knowl. Econ.* **2025**, *16*, 9624–9648. [CrossRef]
84. Otika, U.S.; Ejiofor, U.H.; Olise, C.M. Technological-Behavioral-Environmental (TPE) Framework; a Conceptual Model for Consumers Acceptance of Digital Marketing Devices. In *Technological, Pandemic Disruptions, and Management Sciences' Theory and Practice: Challenges, Responses, and Strategic Choices*; The Academy of Management Nigeria: Abuja, Nigeria, 2021; p. 955.
85. Sastararuji, D.; Hoonsopon, D.; Pitchayadol, P.; Chiwamit, P. Cloud Accounting Adoption in Small and Medium Enterprises An Integrated Conceptual Framework Five factors of determinant were identified by integrated Technology-Organization-Environment (TOE) framework, Diffusion of Innovation (DOI), Institutional Theory. In Proceedings of the 2021 The 2nd International Conference on Industrial Engineering and Industrial Management, Barcelona, Spain, 8–11 January 2021; pp. 32–38.

86. Kripfganz, S.; Sarafidis, V. Instrumental-variable estimation of large-T panel-data models with common factors. *Stata J.* **2021**, *21*, 659–686. [CrossRef]
87. Meza, I.; Singh, R. Nested nonparametric instrumental variable regression: Long term, mediated, and time varying treatment effects. *arXiv* **2021**, arXiv:2112.14249.
88. Ao, X.; Zhang, Y.; Zhou, Y.; Xu, D. Research on weighted cluster analysis method of panel data. *J. Phys. Conf. Ser.* **2021**, *1848*, 012036. [CrossRef]
89. Xun, J.; Li, W.; Xu, Y. A Random Matrix Model of Regional Economic Disparity Based on Spatial Panel Data Analysis. *Math. Probl. Eng.* **2022**, *2022*, 7993818. [CrossRef]
90. Benecchi, A.; Bottoni, C.; Ciapanna, E.; Frigo, A.; Milan, A.; Scarinzi, E. Digitalisation in Italy: Evidence from a new regional index. *Soc. Indic. Res.* **2023**, *169*, 23–54. [CrossRef]
91. Rella, A.; Schiuma, G.; Raimo, N.; Ricciardelli, A.; Vitolla, F. Driving Digital Transformation: Investigating the Key Determi-nants of Smart Technology Adoption in Italian Municipalities. *Int. J. Public Adm.* **2025**, 1–14. [CrossRef]
92. Capello, R.; Lenzi, C.; Panzera, E. The nexus between the digital service economy and intraregional wage inequalities. *Econ. Geogr.* **2024**, *100*, 246–273. [CrossRef]
93. Prota, F.; Viesti, G. *Linking the 'Recovery and Resilience Plan' and Smart Specialisation: The Italian Case*; Joint Research Centre: Seville, Spain, 2022.
94. Marucci, A.; Zullo, F.; Fiorini, L.; Saganeiti, L. *Innovation in Urban and Regional Planning*; Proceedings of INPUT; Springer: Berlin/Heidelberg, Germany, 2023; Volume 2.
95. Di Stefano, V.; Paletto, A.; Cortignani, R.; Di Domenico, G. Fostering Digitalization: How Local Policies Are Transforming Rural Areas in Italy. *Forests* **2025**, *16*, 260. [CrossRef]
96. Costa, S.; De Santis, S.; Monducci, R. Reacting to the COVID-19 crisis: State, strategies and perspectives of Italian firms. *Riv. Di Stat. Uff./Rev. Off. Stat.* **2022**, *1*, 73–107.
97. Crupi, A.; Del Sarto, N.; Di Minin, A.; Gregori, G.L.; Lepore, D.; Marinelli, L.; Spigarelli, F. The digital transformation of SMEs–a new knowledge broker called the digital innovation hub. *J. Knowl. Manag.* **2020**, *24*, 1263–1288. [CrossRef]
98. Granata, G. How Italian Companies Changed Their Communication Strategies During the COVID-19 Pandemic. *Int. J. Acad. Res. Bus. Soc. Sci.* **2021**, *11*, 97–106. [CrossRef]
99. Wang, L.; He, Y.; Wu, R. Digitization meets energy transition: Shaping the future of environmental sustainability. *Energies* **2024**, *17*, 767. [CrossRef]
100. Cascini, F.; A Causio, F.; Melnyk, A. Monitoring the digital healthcare transition in Italy: Initiatives, metrics and feedback from Primary Care Physicians. *Eur. J. Public Health* **2024**, *34*, ckae144-074. [CrossRef]
101. Melo, I.C.; Queiroz, G.A.; Junior, P.N.A.; de Sousa, T.B.; Yushimito, W.F.; Pereira, J. Sustainable digital transformation in small and medium enterprises (SMEs): A review on performance. *Heliyon* **2023**, *9*, e13908. [CrossRef]
102. Ardito, L.; Raby, S.; Albino, V.; Bertoldi, B. The duality of digital and environmental orientations in the context of SMEs: Im-plications for innovation performance. *J. Bus. Res.* **2021**, *123*, 44–56. [CrossRef]
103. Mola, L.; Roffia, P. Digitalizing sales channels in wine business SMEs: The role of internal and external factors between oppor-tunities and risks. *Br. Food J.* **2025**, *127*, 1395–1419. [CrossRef]
104. Lai, H.; Huang, T.; Lu, B.; Zhang, S.; Xiaog, R. Silhouette coefficient-based weighting k-means algorithm. *Neural Comput. Appl.* **2025**, *37*, 3061–3075. [CrossRef]
105. Panati, H.R.; Joy, E. Propensity modelling for predicting customer behaviour using clustering techniques for effective business analysis. In *Proceedings of the 2nd International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN)*; IEEE: Piscataway, NJ, USA, 2023; pp. 1–6.
106. Aziz, S.M.; Rifai, N.A.K. Pengelompokkan Ekspor Kopi Menurut Negara Tujuan Menggunakan Metode K-Means Clustering dengan Silhouette Coefficient. *Bdg. Conf. Ser. Stat.* **2022**, *2*, 416–424.
107. Utami, I.T.; Suryaningrum, F.; Ispriyanti, D. K-means cluster count optimization with silhouette index validation and davies bouldin index (CASE study: Coverage of pregnant women, childbirth, and postpartum health services in indonesia in 2020). *BAREKENG J. Ilmu Mat. Terap.* **2023**, *17*, 707–716. [CrossRef]
108. Xu, L.; Ren, T.H.; Zhu, J. K-means clustering-based young users' kitchen needs analysis and personalized design strategy. *Interdiscip. Nurs. Res.* **2024**, *3*, 141–148. [CrossRef]
109. Erda, G.; Gunawan, C.; Erda, Z. Grouping of poverty in Indonesia using K-means with silhouette coefficient. *Parameter J. Stat.* **2023**, *3*, 1–6. [CrossRef]
110. Pradana, C.; Kusumawardani, S.S.; Permanasari, A.E. Comparison clustering performance based on moodle log mining. In *IOP Conference Series: Materials Science and Engineering*; IOP Publishing: Bristol, UK, 2020; Volume 722, p. 012012.
111. Serrecchia, M. Analysis of Internet development and internal digital divide by using the ". it" domain names as an indicator: Evidence from Italy. *Telecommun. Policy* **2024**, *48*, 102870. [CrossRef]

112. Soriano-Miras, R.M.; Rodríguez, F.B.; Mahmud, B.; Gozzo, S.M.A. The world in one click: The digital divide associated with e-commerce in southern Europe. In *Social Problems in Southern Europe*; Edward Elgar Publishing: Cheltenham, UK, 2020; pp. 130–142.
113. Sciarretta, A. Dialectometry-based classification of the Central–Southern Italian dialects. *J. Linguist. Geogr.* **2024**, *12*, 13–23. [CrossRef]
114. Lombardini, S. Italian regional econometric model. *Pap. Reg. Sci.* **2024**, *103*, 100060. [CrossRef]
115. Giannini, M.; Martini, B.; Fiorelli, C. How does firms' broadband adoption affect regional TFP in Italy? *Econ. Politica* **2022**, *40*, 1025–1050. [CrossRef]
116. Deshmukh, V.K.; Rajput, M.S.; Narang, H.K. Performance evaluation and comparison of machine learning algorithms for prediction of electrodeposited copper ions. *Adv. Mater. Process. Technol.* **2024**, *10*, 1282–1292. [CrossRef]
117. Papakonstantinou, M.; Zoulias, E.; Gallos, P.; Mantas, J. Development of a Decision Support System for Predicting the Evolution of Epidemics Using Open-Source Software Tools. *Stud. Health Technol. Inform.* **2025**, *323*, 429–433. [PubMed]
118. Yugay, O.; Yerashenia, N.; Budimir, D. Study of Machine Learning Models for IoT Based Efficient Classroom Usage. In *International Conference on Next Generation Wired/Wireless Networking*; Springer Nature: Cham, Switzerland, 2023; pp. 253–260.
119. Yoon, C.; Kim, J.; Kim, H.-J.; Kim, C.S.; Choi, S.W.; Lee, I.-K. Comparison of Machine Learning Algorithms for Propagation Path Loss Prediction Using Ray Tracing Data. In *Proceedings of the 2024 15th International Conference on Information and Communication Technology Convergence (ICTC)*; IEEE: Piscataway, NJ, USA, 2024; pp. 2177–2179.
120. Farmer, M. The Relationship Between Internet Adoption and Self-Reported Diabetes Prevalence in United States Counties. Ph.D. Dissertation, University of Missouri-Columbia, Columbia, MO, USA, 2024.
121. Pratamansyah, S.R. Transformasi digital dan pertumbuhan UMKM: Analisis dampak teknologi pada kinerja usaha kecil dan menengah di Indonesia. *J. Akunt. Manaj. Dan Perenc. Kebijak.* **2024**, *2*, 17. [CrossRef]
122. Li, X.; Zheng, L. Consumers adoption behavior prediction through technology acceptance model and machine learning mod-els. In *Statistics for Data Science and Policy Analysis*; Springer: Singapore, 2020; pp. 333–346.
123. Akhtiamov, D.; Ghane, R.; Hassibi, B. Regularized linear regression for binary classification. In *IEEE International Symposium on Information Theory (ISIT)*; IEEE: Piscataway, NJ, USA, 2024; pp. 202–207.
124. Paccoud, I.; Baumann, M.; Le Bihan, E.; Pétré, B.; Breinbauer, M.; Böhme, P.; Chauvel, L.; Leist, A.K. Socioeconomic and behavioural factors associated with access to and use of personal health records. *BMC Med. Inform. Decis. Mak.* **2021**, *21*, 18. [CrossRef]
125. Ndofirepi, T.M.; Steyn, R. Unravelling the Digital Thread: How Access, Protection, and Adoption Drive Technological Entrepreneurship. *Adm. Sci.* **2024**, *14*, 185. [CrossRef]
126. Jamil, W.; Bouchachia, A. Competitive regularised regression. *Neurocomputing* **2020**, *390*, 374–383. [CrossRef]
127. Fang, X.; Xu, Y.; Li, X.; Lai, Z.; Wong, W.K.; Fang, B. Regularized label relaxation linear regression. *IEEE Trans. Neural Netw. Learn. Syst.* **2017**, *29*, 1006–1018. [CrossRef] [PubMed]
128. Shikhman, V.; Müller, D. *Mathematical Foundations of Big Data Analytics*; Springer: Wiesbaden, Germany, 2021.
129. Phillips, J.M. *Mathematical Foundations for Data Analysis*; Springer: Cham, Switzerland, 2021.
130. Mhlanga, D.; Dunga, H. Demand for internet services before and during the Covid-19 pandemic: What lessons are we learn-ing in South Africa? *Int. J. Res. Bus. Soc. Sci.* **2023**, *12*, 626–640. [CrossRef]
131. Eshov, M.; Ismailova, N.; Kuldasheva, Z. International Trade and Economic Growth in Central Asia: The Role of Digital Economy. In Proceedings of the 7th International Conference on Future Networks and Distributed Systems, Dubai, United Arab Emirates, 21–23 December 2023; pp. 579–584.
132. Mitchell, C. *Broadband at the Speed of Light: How Three Communities Built Next-Generation Networks*; Institute for Local Self-Reliance: Washington, DC, USA, 2012.
133. Manmadhan, M.; Kurian, G.S.; Ravi, N. Challenges in Democratizing E-commerce in India through Open Network for Digital Commerce: A Seller Network Node Perspective. In *Proceedings of the 15th International Conference on Computing Communication and Net-Working Technologies (ICCCNT)*; IEEE: Piscataway, NJ, USA, 2024; pp. 1–8.
134. Nesterov, A.I. On clustering coefficients in complex networks. *arXiv* **2024**, arXiv:2401.02999. [CrossRef]
135. Liu, J.; Pan, B.; Zhang, X.; Li, D. Mobile e-commerce information system based on industry cluster under edge computing. *Mob. Inf. Syst.* **2021**, *2021*, 7930799. [CrossRef]
136. Li, B.; Lei, Q. Hybrid IoT and data fusion model for e-commerce big data analysis. *Wirel. Commun. Mob. Comput.* **2022**, *2022*, 2292321. [CrossRef]
137. Aguilar-Alarcón, J.J.; Hernández-Gómez, J.C.; Romero-Valencia, J. The clustering coefficient for graph products. *Axioms* **2023**, *12*, 968. [CrossRef]
138. Ermagun, A.; Levinson, D. An introduction to the network weight matrix. *Geogr. Anal.* **2018**, *50*, 76–96. [CrossRef]
139. Huda, N.M.; Imro'aH, N. Determination of the best weight matrix for the Generalized Space Time Autoregressive (GSTAR) model in the Covid-19 case on Java Island Indonesia. *Spat. Stat.* **2023**, *54*, 100734. [CrossRef]

140. Costantini, A.; Rubini, S. Workplace innovation through proactivity: Job crafting and work engagement dynamics among remote workers in Italy. In *A Research Agenda for Workplace Innovation: The Challenge of Disruptive Transitions*; Edward Elgar Publishinbg Ltd.: Cheltenham, UK, 2021.
141. Rizzica, L.; Roma, G.; Rovigatti, G. The Effects of Deregulating Retail Operating Hours: Empirical Evidence from Italy. *J. Law Econ.* **2023**, *66*, 21–52. [CrossRef]
142. Irie, K.; Schlag, I.; Csordás, R.; Schmidhuber, J. A modern self-referential weight matrix that learns to modify itself. In *International Conference on Machine Learning*; PMLR: Oxford, UK, 2022; pp. 9660–9677.
143. Wang, Y.; Tian, Y.; Nasrullah, M.; Zhang, R. Does social capital influence farmers'e-commerce entrepreneurship? China's regional evidence. *Electron. Commer. Res.* **2024**, *25*, 3397–3425. [CrossRef]
144. Rahmatullah. Penafsiran Hukum Terhadap Objek Penguasaan Negara Tentang Sumber Daya Ekonomi Digital di Indonesia. Ph.D. Dissertation, UNS Sebelas Maret University, Surakarta, Indonesia, 2025.
145. Restrepo-Morales, J.A.; Valencia-Cárdenas, M.; García-Pérez-De-Lema, D. The role of technological innovation in the mitigation of the crisis generated by COVID-19: An empirical study of small and medium-sized businesses (SMEs) in Latin America. *Int. Stud. Manag. Organ.* **2024**, *54*, 120–136. [CrossRef]
146. Bramoullé, Y.; Genicot, G. Diffusion and targeting centrality. *J. Econ. Theory* **2024**, *222*, 105920. [CrossRef]
147. Janyasupab, P.; Suratanee, A.; Plaimas, K. Network diffusion with centrality measures to identify disease-related genes. *Math. Biosci. Eng.* **2021**, *18*, 2909–2929. [CrossRef]
148. Sciabolazza, V.L.; Riccetti, L. Diffusion delay centrality: Decelerating diffusion processes across networks. *Ind. Corp. Change* **2022**, *31*, 980–1003. [CrossRef]
149. Bai, Y.; Zhang, Z.; Chen, T.; Peng, W. Analyst network centrality, forecast accuracy, and persistent influence. *Appl. Econ.* **2024**, *56*, 6667–6689. [CrossRef]
150. Jha, A. Redefining Network Topology in Complex Systems: Merging Centrality Metrics, Spectral Theory, and Diffusion Dynamics. *arXiv* **2025**, arXiv:2503.21709. [CrossRef]
151. Zhou, B.; Yan, X.; Lv, Y.; Xuan, Q. Adversarial attacks on clustering coefficient in complex networks. *IEEE Trans. Circuits Syst. II Express Briefs* **2023**, *71*, 2199–2203. [CrossRef]
152. Rodríguez-Méndez, V.; Ser-Giacomi, E.; Hernández-García, E. Clustering coefficient and periodic orbits in flow networks. *Chaos Interdiscip. J. Nonlinear Sci.* **2017**, *27*, 035803. [CrossRef]
153. Kooij, R.E.; Sørensen, N.H.; Bouffanais, R. Tuning the clustering coefficient of generalized circulant networks. *Phys. A Stat. Mech. Its Appl.* **2021**, *578*, 126088. [CrossRef]
154. Kalna, G.; Higham, D.J. Clustering coefficients for weighted networks. In Proceedings of the Symposium on Network Analysis in Natural Sciences and Engineering, Bristol, UK, 15–18 August 2006; Volume 45.
155. Carta, F.; Straccamore, I. Digitalizzazione e mercato del lavoro: Uno studio sulle disparità territoriali nel Mezzogiorno. *Riv. Econ. Mezzog.* **2024**, *38*, 803–841.
156. Bruno, G.; Diglio, A.; Piccolo, C.; Pipicelli, E. A reduced composite indicator for digital divide measurement at the regional lev-el: An application to the digital economy and society index (DESI). *Technol. Forecast. Soc. Change* **2023**, *190*, 122461. [CrossRef]
157. Zeinali, Z.; Mussari, R. Expansion of the digital realm: Analyzing DESI and digital transformation across EU member states, with a focus on Spain and Italy: A cluster analysis. *Public Organ. Rev.* **2025**, *25*, 1233–1273. [CrossRef]
158. Russo, F. Il ruolo dell'AgID nella digitalizzazione della pubblica amministrazione: Considerazioni intorno al piano triennale per l'informatica 2024–2026. *Riv. Sci. Trimest. Dirit. Amminstrativo* **2025**, *3*, 1365–1387.
159. De Donno, M. Le pubbliche amministrazioni come ecosistemi digitali: Regole e prassi di una trasformazione organizzativa. *Munus Riv. Giuridica Serv. Pubblici* **2025**, *2*, 225–254.
160. Scotti, F.; Caporali, C.; Luca, D. NextGenerationEU lands: Assessing the territorial allocation of Italy's recovery and resilience funds. *SSRN Electron. J.* **2025**, 1–58. [CrossRef]
161. Arduini, G.; Chiusaroli, D. Le TIC tra digital divide, universal design e accessibilità. *Ital. J. Spec. Educ. Incl.* **2023**, *11*, 84–89.
162. Falcone, M. La data science come nuovo sapere per governare le intelligenze artificiali nelle pubbliche amministrazioni. *Riv. Ital. Inform. Dirit.* **2024**, *6*, 527–544.
163. Nisticò, R.; Prota, F. Sviluppo industriale per poli? Le zone economiche speciali in Italia e nel PNRR. *Stato Mercato* **2023**, *43*, 227–264.
164. Servidio, G. Obiettivi e strumenti della politica industriale: Un'analisi a livello territoriale. *Riv. Econ. Mezzog.* **2023**, *37*, 25–76.
165. Rasyidan, I.T.; Fachira, I. Digital Marketing Strategy Formulation to Increase Engagement in Course and Training Institution Industry (Case Study: PT Karisma Garuda Mulia). *Int. J. Curr. Sci. Res. Rev.* **2023**, *6*, 3688–3703. [CrossRef]
166. Kumar, M.; Pullman, M.; Bouzdine-Chameeva, T.; Sanchez Rodrigues, V. The role of the hub-firm in developing innovation capabilities: Considering the French wine industry cluster from a resource orchestration lens. *Int. J. Oper. Prod. Manag.* **2022**, *42*, 526–551. [CrossRef]

167. Ollerenshaw, J.; Corbett, J.; Thompson, H. Increasing the digital literacy skills of regional SMEs through high-speed broad-band access. *Small Enterp. Res.* **2021**, *28*, 115–133. [CrossRef]
168. Nazir, M.A.; Roomi, M.A. Barriers to adopting electronic commerce for small and medium-sized enterprises in emerging economies. *EMAJ Emerg. Mark. J.* **2020**, *1*, 43–55. [CrossRef]
169. Andika, P.B.; Rashid, A.; Tri Abriana, M.R. Unincorporated Merchant: How Does the Law Protect Consumers in Electronic Commerce? *Subst. Justice Int. J. Law* **2023**, *6*, 71–82.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.